

```

pmax = 115*ones(1,100);
pmin = -145*ones(1,100);
semilogx(w,phase,'o',w,phase,'-',w,pmax,'--i',w,pmin,':i')

% ***** Enter grid, xlabel, and ylabel *****

grid
xlabel('Frequency (rad/sec)')
ylabel('Phase deg')

```

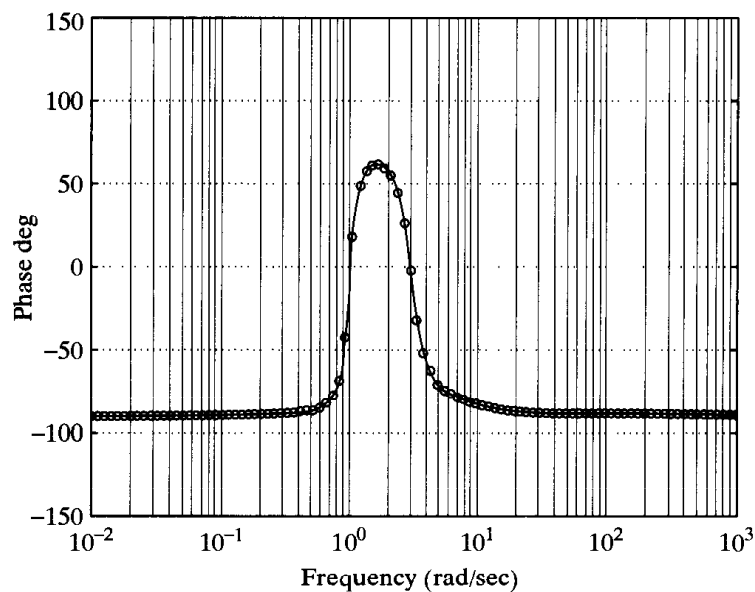
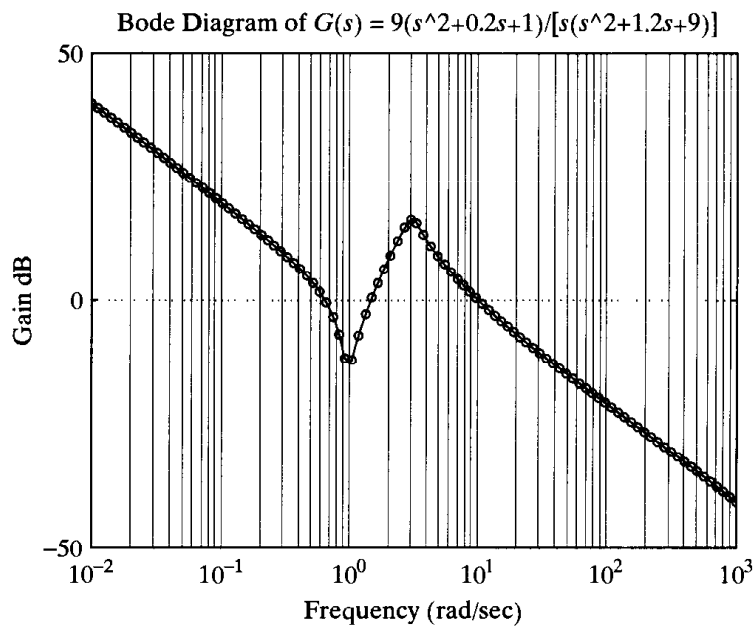


Figure 8–23
 Bode diagram
 of $G(s)$

$$= \frac{9(s^2 + 0.2s + 1)}{s(s^2 + 1.2s + 9)}$$

EXAMPLE 8-5

Consider a system with the following open-loop transfer function:

$$G(s) = \frac{1}{s^2 + 1}$$

This open-loop transfer function has poles on the $j\omega$ axis at $\pm j$.

MATLAB Program 8-5 may be used to plot the Bode diagram for this system. The resulting plot is shown in Figure 8-24. Theoretically, the magnitude becomes infinite at a frequency point where $\omega = 1$ rad/sec. However, this frequency point is not among the computing frequency points. In the plot the peak magnitude is shown to be approximately 50 dB. This value is computed near, but not exactly at, $\omega = 1$ rad/sec.

MATLAB Program 8-5

```
num = [0 0 1];
den = [1 0 1];
bode(num,den)
subplot(2,1,1);
title('Bode Diagram of G(s) = 1/(s^2 + 1)')
```

If, however, one of the computing frequency points coincides with the pole at $\omega = 1$, then the magnitude becomes infinite at this point. MATLAB sends out warning messages. See MATLAB

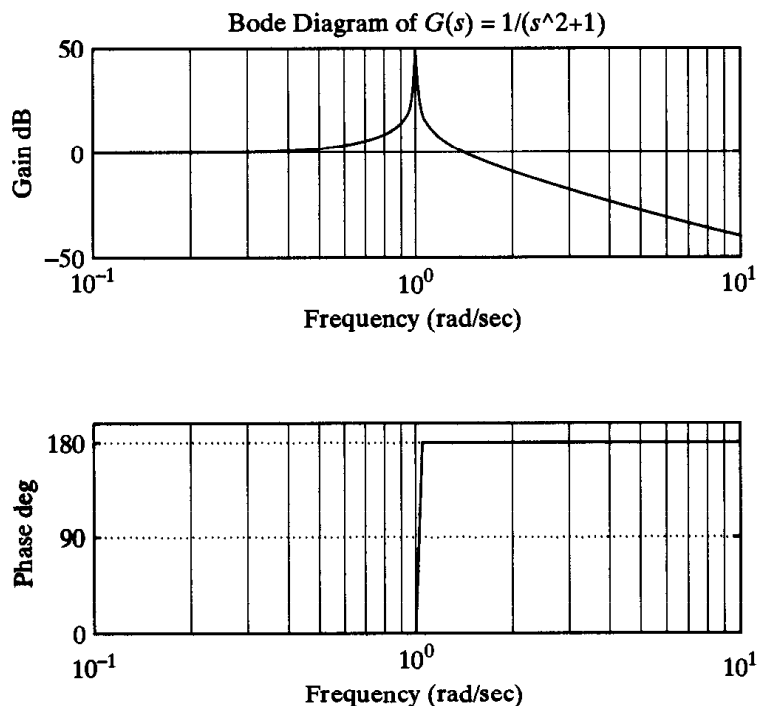


Figure 8-24
Bode diagram of
 $G(s) = \frac{1}{s^2 + 1}$.

Program 8-6 where computing points include the point at $\omega = 1$ rad/sec. (There are 101 computing points in this case. The computing points range from $\omega = 0.1$ to $\omega = 10$. The fifty-first point is at $\omega = 1$.) When MATLAB Program 8-6 is entered into the computer, warning messages appear, as shown. The resulting Bode diagram, shown in Figure 8-25, does not include the computed magnitude at $\omega = 1$. (Theoretically, this magnitude is infinite.) The magnitude curve shows the peak value at about 20 dB. The phase curve shows a gradual change in the phase angle from 0° to $+180^\circ$ near point $\omega = 1$. (Theoretically, the change in the phase angle from 0° to $+180^\circ$ should be abrupt at $\omega = 1$.) Obviously, the Bode diagram shown in Figure 8-25 is incorrect.

If the w vector contains such a frequency point, where the gain becomes infinite, change the number of frequency points, for example, from 101 to 100. Normally, a small change in the number of frequency points will avoid this kind of problem.

MATLAB Program 8-6
<pre> num = [0 0 1]; den = [1 0 1]; w = logspace(-1,1,101); bode(num,den,w) </pre>
Warning: Divide by zero subplot(2,1,1); title('Incorrect Bode Diagram')

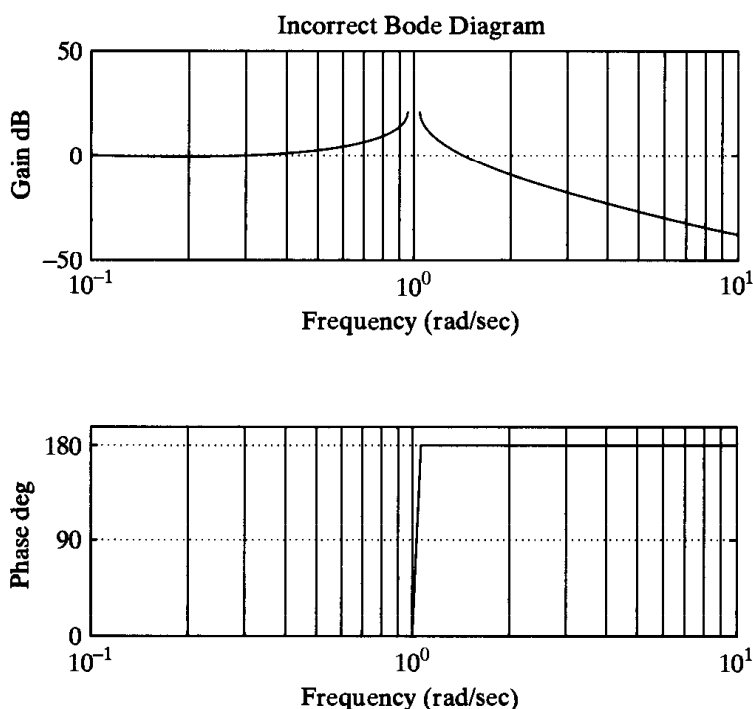


Figure 8-25
Incorrect Bode
diagram of
 $G(s) = \frac{1}{s^2 + 1}$.

Obtaining Bode diagrams of systems defined in state space. Consider the system defined by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

where $\mathbf{x}$ = state vector (n -vector)

$\mathbf{y}$ = output vector (m -vector)

$\mathbf{u}$ = control vector (r -vector)

$\mathbf{A}$ = state matrix ($n \times n$ matrix)

$\mathbf{B}$ = control matrix ($n \times r$ matrix)

$\mathbf{C}$ = output matrix ($m \times n$ matrix)

$\mathbf{D}$ = direct transmission matrix ($m \times r$ matrix)

A Bode diagram for this system may be obtained by entering the command

bode(A,B,C,D)

or

bode(A,B,C,D,iu)

The command bode(A,B,C,D) produces a series of Bode plots, one for each input of the system, with the frequency range automatically determined. (More points are used when the response is changing rapidly.)

The command bode(A,B,C,D,iu) where iu is the i th input of the system, produces the Bode diagrams from the input iu to all the outputs ($y_1, y_2, \dots, y_m$) of the system, with frequency range automatically determined. (The scalar iu is an index into the inputs of the system and specifies which input is to be used for plotting Bode diagrams). If the control vector $\mathbf{u}$ has three inputs such that

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

then iu must be set to either 1, 2, or 3.

If the system has only one input u , then either of the following commands may be used:

bode(A,B,C,D)

or

bode(A,B,C,D,1)

EXAMPLE 8-6

Consider the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -25 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 25 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

This system has one input u and one output y . By using the command

`bode(A,B,C,D)`

and entering MATLAB Program 8-7 into the computer, we obtain the Bode diagram shown in Figure 8-26.

MATLAB Program 8-7
<pre>A = [0 1;-25 -4]; B = [0;25]; C = [1 0]; D = [0]; bode(A,B,C,D) subplot(2,1,1); title('Bode Diagram')</pre>

If we replace the command `bode(A,B,C,D)` in MATLAB Program 8-7 with

`bode(A,B,C,D,1)`

then MATLAB will produce the Bode diagram identical to that shown in Figure 8-26.

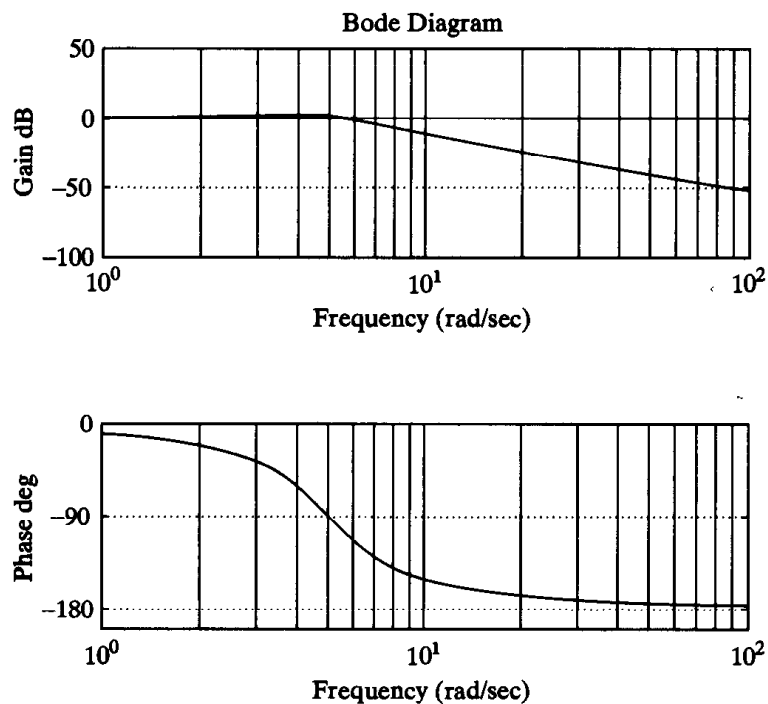


Figure 8-26
Bode diagram of the
system considered in
Example 8-6.

Note that if we use, by mistake, the command

```
bode(A,B,C,D,2)
```

MATLAB produces an error message, because the present system has only one input and it should be set to '1', not '2' or any other number.

8-4 POLAR PLOTS

The polar plot of a sinusoidal transfer function $G(j\omega)$ is a plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ on polar coordinates as ω is varied from zero to infinity. Thus, the polar plot is the locus of vectors $|G(j\omega)| \angle G(j\omega)$ as ω is varied from zero to infinity. Note that in polar plots a positive (negative) phase angle is measured counter-clockwise (clockwise) from the positive real axis. The polar plot is often called the Nyquist plot. An example of such a plot is shown in Figure 8-27. Each point on the polar plot of $G(j\omega)$ represents the terminal point of a vector at a particular value of ω . In the polar plot, it is important to show the frequency graduation of the locus. The projections of $G(j\omega)$ on the real and imaginary axes are its real and imaginary components. Both the magnitude $|G(j\omega)|$ and phase angle $\angle G(j\omega)$ must be calculated directly for each frequency ω in order to construct polar plots. Since the logarithmic plot is easy to construct, however, the data necessary for plotting the polar plot may be obtained directly from the logarithmic plot if the latter is drawn first and decibels are converted into ordinary magnitude. Or, of course, MATLAB may be used to obtain a polar plot $G(j\omega)$ or to obtain $|G(j\omega)|$ and $\angle G(j\omega)$ accurately for various values of ω in the frequency range of interest. (See Section 8-5.)

An advantage in using a polar plot is that it depicts the frequency-response characteristics of a system over the entire frequency range in a single plot. One disadvantage is that the plot does not clearly indicate the contributions of each individual factor of the open-loop transfer function.

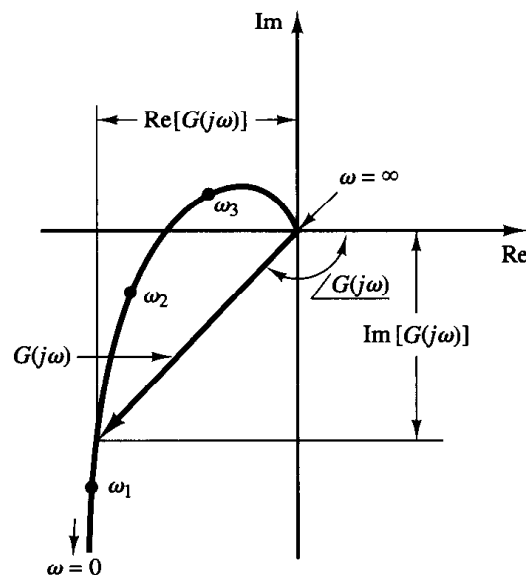


Figure 8-27
Polar plot.

Integral and derivative factors $(j\omega)^{\mp 1}$. The polar plot of $G(j\omega) = 1/j\omega$ is the negative imaginary axis since

$$G(j\omega) = \frac{1}{j\omega} = -j \frac{1}{\omega} = \frac{1}{\omega} \angle -90^\circ$$

The polar plot of $G(j\omega) = j\omega$ is the positive imaginary axis.

First-order factors $(1 + j\omega T)^{\mp 1}$. For the sinusoidal transfer function

$$G(j\omega) = \frac{1}{1 + j\omega T} = \frac{1}{\sqrt{1 + \omega^2 T^2}} \angle -\tan^{-1} \omega T$$

the values of $G(j\omega)$ at $\omega = 0$ and $\omega = 1/T$ are, respectively,

$$G(j0) = 1 \angle 0^\circ \quad \text{and} \quad G\left(j \frac{1}{T}\right) = \frac{1}{\sqrt{2}} \angle -45^\circ$$

If ω approaches infinity, the magnitude of $G(j\omega)$ approaches zero and the phase angle approaches -90° . The polar plot of this transfer function is a semicircle as the frequency ω is varied from zero to infinity, as shown in Figure 8-28(a). The center is located at 0.5 on the real axis, and the radius is equal to 0.5.

To prove that the polar plot is a semicircle, define

$$G(j\omega) = X + jY$$

where

$$X = \frac{1}{1 + \omega^2 T^2} = \text{real part of } G(j\omega)$$

$$Y = \frac{-\omega T}{1 + \omega^2 T^2} = \text{imaginary part of } G(j\omega)$$

Then we obtain

$$\left(X - \frac{1}{2}\right)^2 + Y^2 = \left(\frac{1}{2} \frac{1 - \omega^2 T^2}{1 + \omega^2 T^2}\right)^2 + \left(\frac{-\omega T}{1 + \omega^2 T^2}\right)^2 = \left(\frac{1}{2}\right)^2$$

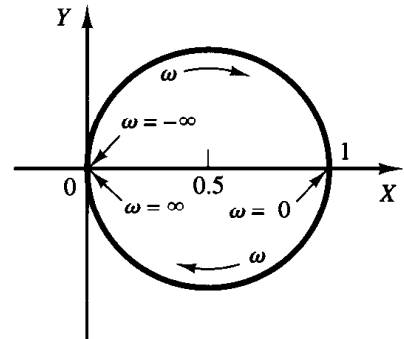
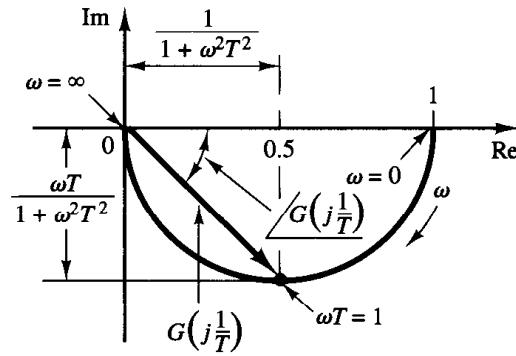


Figure 8-28
(a) Polar plot of $1/(1 + j\omega T)$; (b) plot of $G(j\omega)$ in X - Y plane.

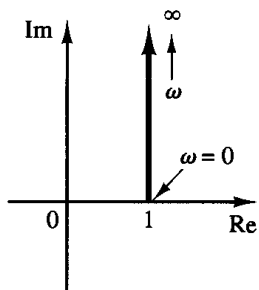


Figure 8-29
Polar plot of $1 + j\omega T$.

Thus, in the X - Y plane $G(j\omega)$ is a circle with center at $X = \frac{1}{2}$, $Y = 0$ and with radius $\frac{1}{2}$, as shown in Figure 8-28(b). The lower semicircle corresponds to $0 \leq \omega \leq \infty$, and the upper semicircle corresponds to $-\infty \leq \omega \leq 0$.

The polar plot of the transfer function $1 + j\omega T$ is simply the upper half of the straight line passing through point $(1,0)$ in the complex plane and parallel to the imaginary axis, as shown in Figure 8-29. The polar plot of $1 + j\omega T$ has an appearance completely different from that of $1/(1 + j\omega T)$.

Quadratic factors $[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]^{-1}$. The low- and high-frequency portions of the polar plot of the following sinusoidal transfer function

$$G(j\omega) = \frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2}, \quad \text{for } \zeta > 0$$

are given, respectively, by

$$\lim_{\omega \rightarrow 0} G(j\omega) = 1 \angle 0^\circ \quad \text{and} \quad \lim_{\omega \rightarrow \infty} G(j\omega) = 0 \angle -180^\circ$$

The polar plot of this sinusoidal transfer function starts at $1 \angle 0^\circ$ and ends at $0 \angle -180^\circ$ as ω increases from zero to infinity. Thus, the high-frequency portion of $G(j\omega)$ is tangent to the negative real axis. The values of $G(j\omega)$ in the frequency range of interest can be calculated directly or by use of the Bode diagram or by use of MATLAB.

Examples of polar plots of the transfer function just considered are shown in Figure 8-30. The exact shape of a polar plot depends on the value of the damping ratio ζ , but the general shape of the plot is the same for both the underdamped case ($1 > \zeta > 0$) and overdamped case ($\zeta > 1$).

For the underdamped case at $\omega = \omega_n$, we have $G(j\omega_n) = 1/(j2\zeta)$, and the phase angle at $\omega = \omega_n$ is -90° . Therefore, it can be seen that the frequency at which the

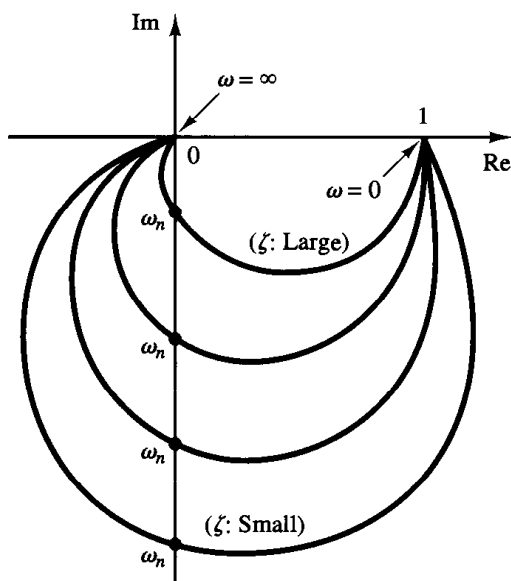


Figure 8-30
Polar plots of
$$\frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2},$$

for $\zeta > 0$.

$G(j\omega)$ locus intersects the imaginary axis is the undamped natural frequency ω_n . In the polar plot, the frequency point whose distance from the origin is maximum corresponds to the resonant frequency ω_r . The peak value of $G(j\omega)$ is obtained as the ratio of the magnitude of the vector at the resonant frequency ω_r to the magnitude of the vector at $\omega = 0$. The resonant frequency ω_r is indicated in the polar plot shown in Figure 8–31.

For the overdamped case, as ζ increases well beyond unity, the $G(j\omega)$ locus approaches a semicircle. This may be seen from the fact that for a heavily damped system the characteristic roots are real and one is much smaller than the other. Since for sufficiently large ζ the effect of the larger root (larger in the absolute value) on the response becomes very small, the system behaves like a first-order one.

Next, consider the following sinusoidal transfer function:

$$\begin{aligned} G(j\omega) &= 1 + 2\zeta \left(j \frac{\omega}{\omega_n} \right) + \left(j \frac{\omega}{\omega_n} \right)^2 \\ &= \left(1 - \frac{\omega^2}{\omega_n^2} \right) + j \left(\frac{2\zeta\omega}{\omega_n} \right) \end{aligned}$$

The low-frequency portion of the curve is

$$\lim_{\omega \rightarrow 0} G(j\omega) = 1 \angle 0^\circ$$

and the high frequency portion is

$$\lim_{\omega \rightarrow \infty} G(j\omega) = \infty \angle 180^\circ$$

Since the imaginary part of $G(j\omega)$ is positive for $\omega > 0$ and is monotonically increasing and the real part of $G(j\omega)$ is monotonically decreasing from unity, the general shape of the polar plot of $G(j\omega)$ is as shown in Figure 8–32. The phase angle is between 0° and 180° .

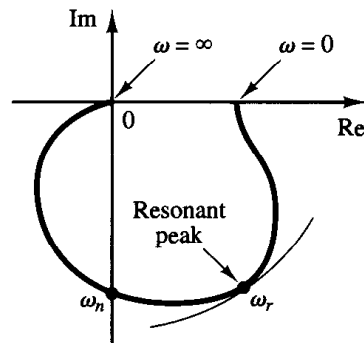


Figure 8–31
Polar plot showing the resonant peak and resonant frequency ω_r .

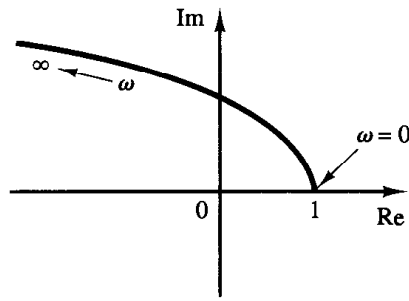


Figure 8-32
Polar plot of $1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2$,
for $\zeta > 0$.

EXAMPLE 8-7

Consider the following second-order transfer function:

$$G(s) = \frac{1}{s(Ts + 1)}$$

Sketch a polar plot of this transfer function.

Since the sinusoidal transfer function can be written

$$G(j\omega) = \frac{1}{j\omega(1 + j\omega T)} = -\frac{T}{1 + \omega^2 T^2} - j\frac{1}{\omega(1 + \omega^2 T^2)}$$

the low-frequency portion of the polar plot becomes

$$\lim_{\omega \rightarrow 0} G(j\omega) = -T - j\infty = \infty \angle -90^\circ$$

and the high-frequency portion becomes

$$\lim_{\omega \rightarrow \infty} G(j\omega) = 0 - j0 = 0 \angle -180^\circ$$

The general shape of the polar plot of $G(j\omega)$ is shown in Figure 8-33. The $G(j\omega)$ plot is asymptotic to the vertical line passing through the point $(-T, 0)$. Since this transfer function involves an integrator ($1/s$), the general shape of the polar plot differs substantially from those of second-order transfer functions that do not have an integrator.

Transport lag. The transport lag

$$G(j\omega) = e^{-j\omega T}$$

can be written

$$G(j\omega) = 1 / \cos \omega T - j \sin \omega T$$

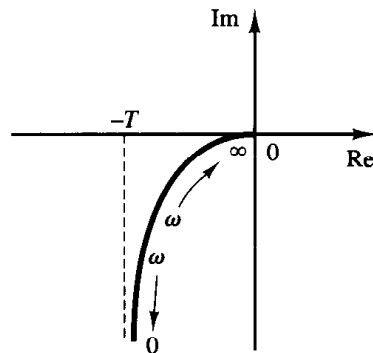


Figure 8-33
Polar plot of $1/[j\omega(1 + j\omega T)]$.

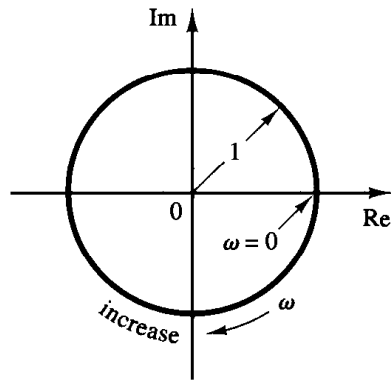


Figure 8-34
Polar plot of transport lag.

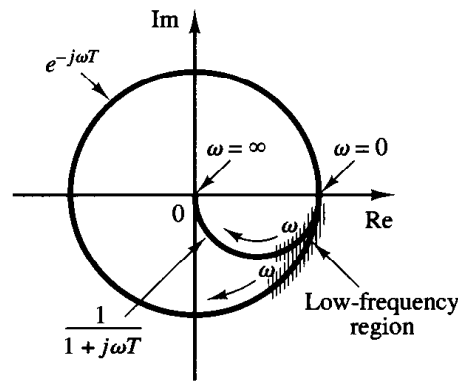


Figure 8-35
Polar plots of $e^{-j\omega T}$ and $1/(1 + j\omega T)$.

Since the magnitude of $G(j\omega)$ is always unity and the phase angle varies linearly with ω , the polar plot of the transport lag is a unit circle, as shown in Figure 8-34.

At low frequencies, the transport lag $e^{-j\omega T}$ and the first-order lag $1/(1 + j\omega T)$ behave similarly, as shown in Figure 8-35. The polar plots of $e^{-j\omega T}$ and $1/(1 + j\omega T)$ are tangent to each other at $\omega = 0$. This may be seen from the fact that, for $\omega \ll 1/T$,

$$e^{-j\omega T} \doteq 1 - j\omega T \quad \text{and} \quad \frac{1}{1 + j\omega T} \doteq 1 - j\omega T$$

For $\omega \gg 1/T$, however, an essential difference exists between $e^{-j\omega T}$ and $1/(1 + j\omega T)$, as may also be seen from Figure 8-35.

EXAMPLE 8-8

Obtain the polar plot of the following transfer function:

$$G(j\omega) = \frac{e^{-j\omega L}}{1 + j\omega T}$$

Since $G(j\omega)$ can be written

$$G(j\omega) = (e^{-j\omega L}) \left(\frac{1}{1 + j\omega T} \right)$$

the magnitude and phase angle are, respectively,

$$|G(j\omega)| = |e^{-j\omega L}| \cdot \left| \frac{1}{1 + j\omega T} \right| = \frac{1}{\sqrt{1 + \omega^2 T^2}}$$

and

$$\angle G(j\omega) = \angle e^{-j\omega L} + \angle \frac{1}{1 + j\omega T} = -\omega L - \tan^{-1} \omega T$$

Since the magnitude decreases from unity monotonically and the phase angle also decreases monotonically and indefinitely, the polar plot of the given transfer function is a spiral, as shown in Figure 8–36.

General shapes of polar plots. The polar plots of a transfer function of the form

$$\begin{aligned} G(j\omega) &= \frac{K(1 + j\omega T_a)(1 + j\omega T_b) \cdots}{(j\omega)^\lambda (1 + j\omega T_1)(1 + j\omega T_2) \cdots} \\ &= \frac{b_0(j\omega)^m + b_1(j\omega)^{m-1} + \cdots}{a_0(j\omega)^n + a_1(j\omega)^{n-1} + \cdots} \end{aligned}$$

where $n > m$ or the degree of the denominator polynomial is greater than that of the numerator, will have the following general shapes:

1. *For $\lambda = 0$ or type 0 systems:* The starting point of the polar plot (which corresponds to $\omega = 0$) is finite and is on the positive real axis. The tangent to the polar plot at $\omega = 0$ is perpendicular to the real axis. The terminal point, which corresponds to $\omega = \infty$, is at the origin, and the curve is tangent to one of the axes.
2. *For $\lambda = 1$ or type 1 systems:* the $j\omega$ term in the denominator contributes -90° to the total phase angle of $G(j\omega)$ for $0 \leq \omega \leq \infty$. At $\omega = 0$, the magnitude of $G(j\omega)$ is infinity, and the phase angle becomes -90° . At low frequencies, the polar plot is asymptotic to a line parallel to the negative imaginary axis. At $\omega = \infty$, the magnitude becomes zero, and the curve converges to the origin and is tangent to one of the axes.
3. *For $\lambda = 2$ or type 2 systems:* The $(j\omega)^2$ term in the denominator contributes -180° to the total phase angle of $G(j\omega)$ for $0 \leq \omega \leq \infty$. At $\omega = 0$, the magnitude of $G(j\omega)$ is infinity, and the phase angle is equal to -180° . At low frequencies, the polar plot is asymptotic to a line parallel to the negative real axis. At $\omega = \infty$, the magnitude becomes zero, and the curve is tangent to one of the axes.

The general shapes of the low-frequency portions of the polar plots of type 0, type 1, and type 2 systems are shown in Figure 8–37. It can be seen that, if the degree of the denominator polynomial of $G(j\omega)$ is greater than that of the numerator, then the $G(j\omega)$

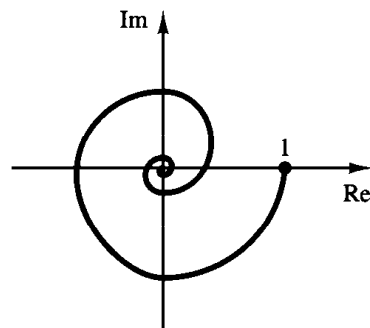


Figure 8–36
Polar plot of $e^{-j\omega L}/(1 + j\omega T)$.

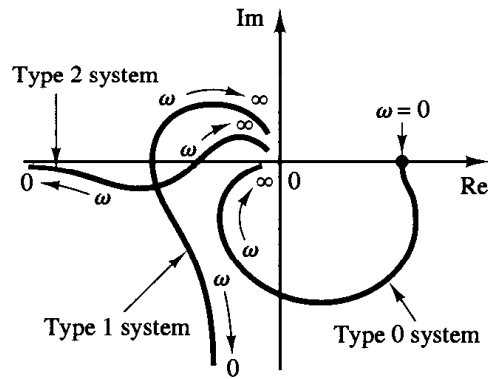
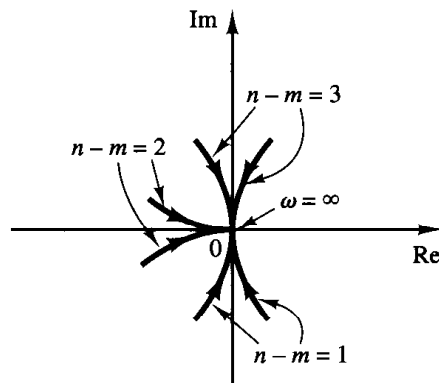


Figure 8-37
Polar plots of type 0, type 1, and type 2 systems.

loci converge to the origin clockwise. At $\omega = \infty$, the loci are tangent to one or the other axes, as shown in Figure 8-38.

Note that any complicated shapes in the polar plot curves are caused by the numerator dynamics, that is, by the time constants in the numerator of the transfer function. Figure 8-39 shows examples of polar plots of transfer functions with numerator dynamics. In analyzing control systems, the polar plot of $G(j\omega)$ in the frequency range of interest must be accurately determined.

Table 8-1 shows sketches of polar plots of several transfer functions.



$$G(j\omega) = \frac{b_o(j\omega)^m + \dots}{a_o(j\omega)^n + \dots}$$

Figure 8-38
Polar plots in the high-frequency range.

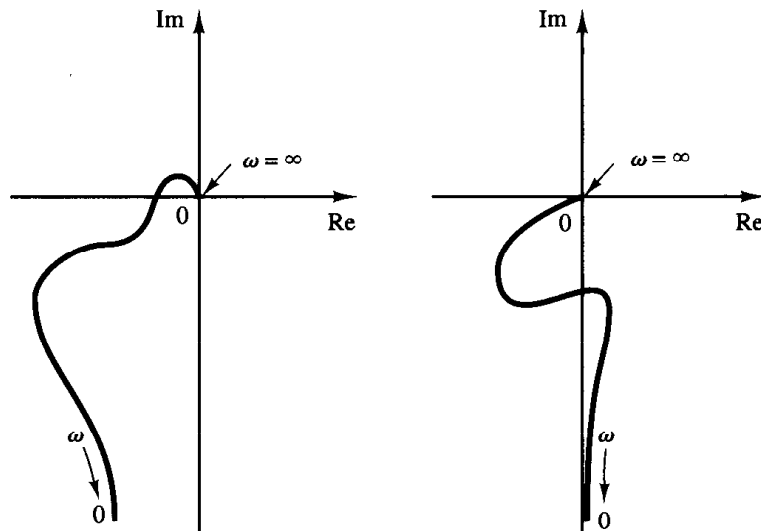


Figure 8-39
Polar plots of transfer functions with numerator dynamics.

Table 8-1 Polar Plots of Simple Transfer Functions

8-5 DRAWING NYQUIST PLOTS WITH MATLAB

Nyquist plots, just like Bode diagrams, are commonly used in the frequency-response representation of linear, time-invariant, feedback control systems. Nyquist plots are polar plots, while Bode diagrams are rectangular plots. One plot or the other may be more convenient for a particular operation, but a given operation can always be carried out in either plot.

The command `nyquist` computes the frequency response for continuous-time, linear, time-invariant systems. When invoked without left-hand arguments, `nyquist` produces a Nyquist plot on the screen.

The command

$$\text{nyquist}(\text{num}, \text{den})$$

draws the Nyquist plot of the transfer function

$$G(s) = \frac{\text{num}(s)}{\text{den}(s)}$$

where `num` and `den` contain the polynomial coefficients in descending powers of s .

The command

$$\text{nyquist}(\text{num}, \text{den}, w)$$

uses the user-specified frequency vector w . The vector w specifies the frequency points in radians per second at which the frequency response will be calculated.

When invoked with the left-hand arguments

$$[\text{re}, \text{im}, w] = \text{nyquist}(\text{num}, \text{den})$$

or

$$[\text{re}, \text{im}, w] = \text{nyquist}(\text{num}, \text{den}, w)$$

MATLAB returns the frequency response of the system in the matrices `re`, `im` and `w`. No plot is drawn on the screen. The matrices `re` and `im` contain the real and imaginary parts of the frequency response of the system evaluated at the frequency points specified in the vector w . Note that `re` and `im` have as many columns as outputs and one row for each element in w .

EXAMPLE 8-9

Consider the following open-loop transfer function:

$$G(s) = \frac{1}{s^2 + 0.8s + 1}$$

Draw a Nyquist plot with MATLAB.

Since the system is given in the form of the transfer function, the command

$$\text{nyquist}(\text{num}, \text{den})$$

may be used to draw a Nyquist plot. MATLAB Program 8-8 produces the Nyquist plot shown in Figure 8-40. In this plot the ranges for the real axis and imaginary axis are automatically determined.

If we wish to draw the Nyquist plot using manually determined ranges, for example, from -2 to 2 on the real axis and from -2 to 2 on the imaginary axis, enter the following command into the computer:

MATLAB Program 8–8

```
num = [0 0 1];
den = [1 0.8 1];
nyquist(num,den)
grid
title('Nyquist Plot of G(s) = 1/(s^2+0.8s+1)')
```

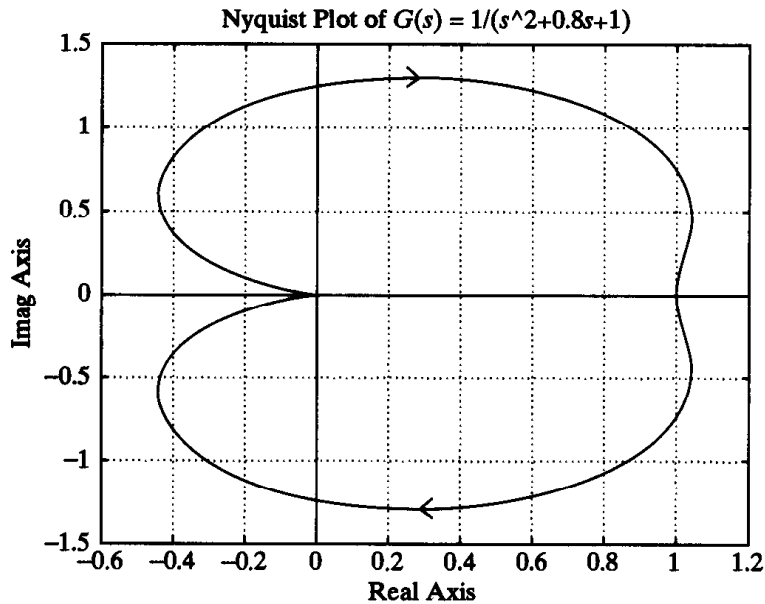


Figure 8–40
Nyquist plot of $G(s)$
$$= \frac{1}{s^2 + 0.8s + 1}$$

```
v = [-2 2 -2 2];
axis(v);
```

or, combining these two lines into one,

```
axis([-2 2 -2 2]);
```

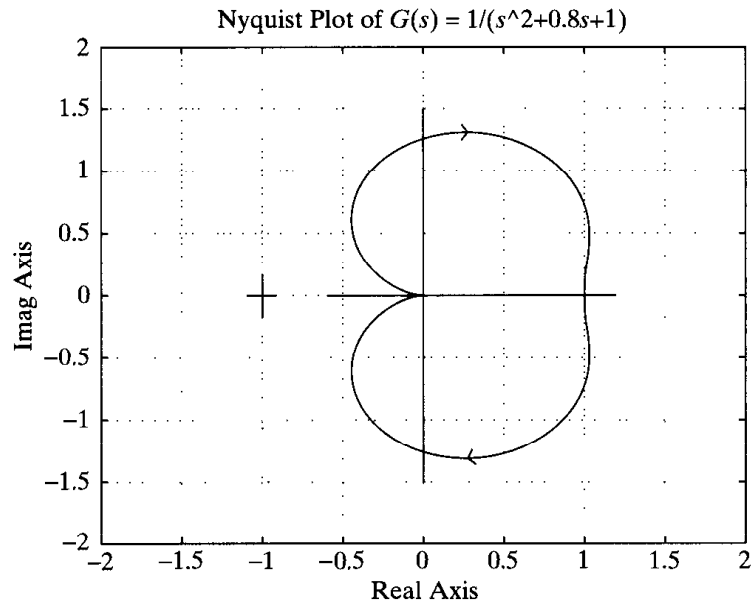
See MATLAB Program 8–9 and the resulting Nyquist plot shown in Figure 8–41.

MATLAB Program 8–9

```
% ----- Nyquist plot -----
num = [0 0 1];
den = [1 0.8 1];
nyquist(num,den)
v = [-2 2 -2 2]; axis(v)
grid
title('Nyquist Plot of G(s) = 1/(s^2 + 0.8s + 1)')
```

Caution. In drawing a Nyquist plot, where MATLAB operation involves “Divide by zero,” the resulting Nyquist plot may be erroneous. For example, if the transfer function $G(s)$ is given by

Figure 8–41
Nyquist plot of $G(s)$
$$= \frac{1}{s^2 + 0.8s + 1}.$$



$$G(s) = \frac{1}{s(s + 1)}$$

then the MATLAB command

```
num = [0 0 1];
den = [1 1 0];
nyquist(num,den)
```

produces an erroneous Nyquist plot. An example of an erroneous Nyquist plot is shown in Figure 8–42. If such an erroneous Nyquist plot appears on the computer, then it can be corrected if we specify the axis(v). For example, if we enter the axis command

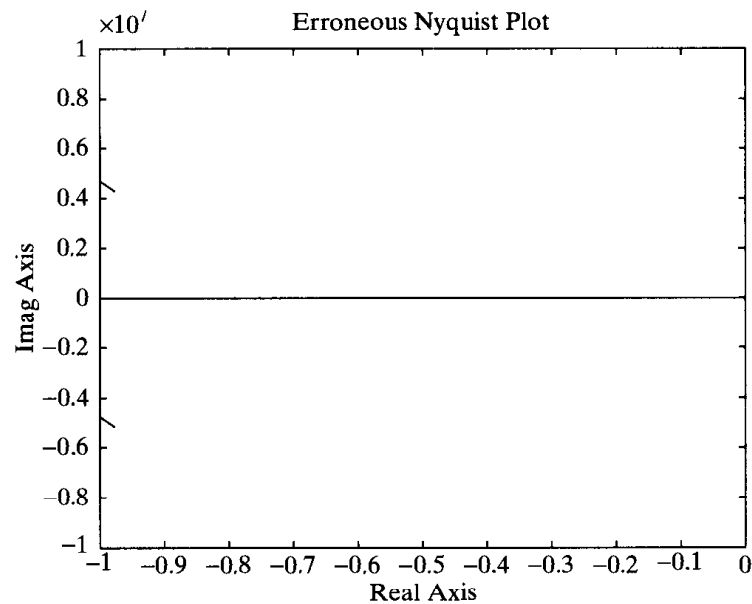


Figure 8–42
Erroneous Nyquist
plot.

$v = [-2 \quad 2 \quad -5 \quad 5]; \text{axis}(v)$

in the computer, then a correct Nyquist plot can be obtained. See Example 8–10.

EXAMPLE 8–10 Draw a Nyquist plot for the following $G(s)$:

$$G(s) = \frac{1}{s(s+1)}$$

MATLAB Program 8–10 will produce a correct Nyquist plot on the computer even though a warning message “Divide by zero” may appear on the screen. The resulting Nyquist plot is shown in Figure 8–43.

MATLAB Program 8–10

```
% ----- Nyquist plot -----  
  
num = [0 0 1];  
den = [1 1 0];  
nyquist(num,den)  
v = [-2 2 -5 5]; axis(v)  
grid  
title('Nyquist Plot of G(s) = 1/[s(s + 1)]')
```

Notice that the Nyquist plot shown in Figure 8–43 includes the loci for both $\omega > 0$ and $\omega < 0$. If we wish to draw the Nyquist plot for only the positive frequency region ($\omega > 0$), then we need to use the command

$[\text{re}, \text{im}, \text{w}] = \text{nyquist}(\text{num}, \text{den}, \text{w})$

A MATLAB program using this nyquist command is shown in MATLAB Program 8–11. The resulting Nyquist plot is presented in Figure 8–44.

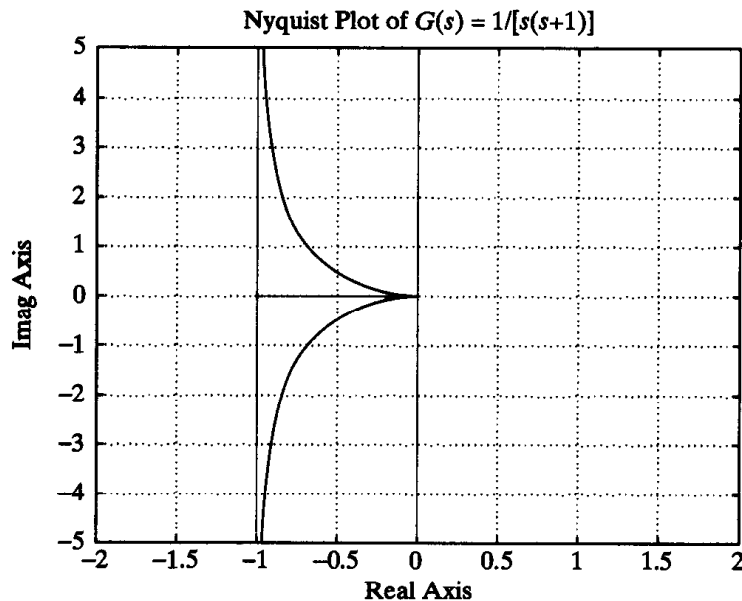


Figure 8–43
Nyquist plot of

$$G(s) = \frac{1}{s(s+1)}$$

MATLAB Program 8-11

```
% ----- Nyquist plot -----

num = [0 0 1];
den = [1 1 0];
w = 0.1:0.1:100;
[re,im,w] = nyquist(num,den,w);
plot(re,im)
v = [-2 2 -5 5]; axis(v)
grid
title('Nyquist Plot of G(s) = 1/[s(s + 1)]')
xlabel('Real Axis')
ylabel('Imag Axis')
```

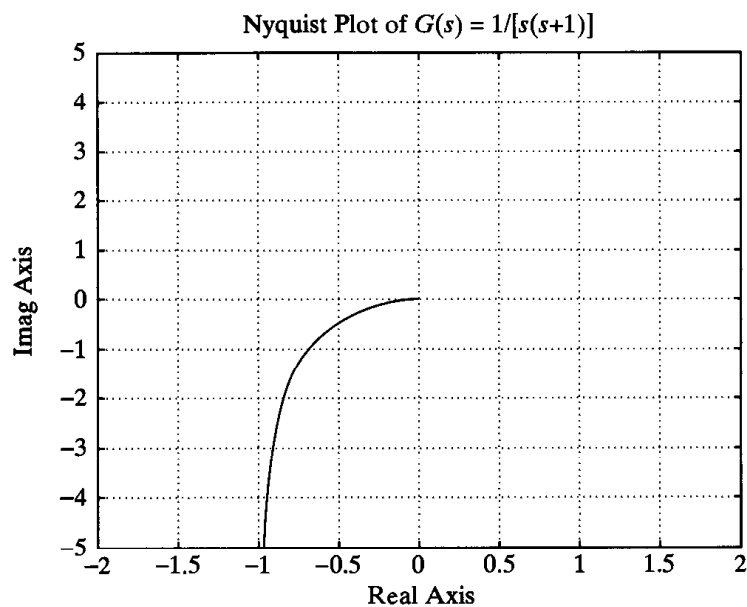


Figure 8-44
Nyquist plot of
 $G(s) = \frac{1}{s(s + 1)}$.

Drawing Nyquist plots of a system defined in state space. Consider the system defined by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

where $\mathbf{x}$ = state vector (n -vector)

$\mathbf{y}$ = output vector (m -vector)

$\mathbf{u}$ = control vector (r -vector)

$\mathbf{A}$ = state matrix ($n \times n$ matrix)

$\mathbf{B}$ = control matrix ($n \times r$ matrix)

$\mathbf{C}$ = output matrix ($m \times n$ matrix)

$\mathbf{D}$ = direct transmission matrix ($m \times r$ matrix)

Nyquist plots for this system may be obtained by use of the command

nyquist(A,B,C,D)

This command produces a series of Nyquist plots, one for each input and output combination of the system. The frequency range is automatically determined.

The command

$$\text{nyquist}(A,B,C,D,iu)$$

produces Nyquist plots from the single input iu to all the outputs of the system, with the frequency range determined automatically. The scalar iu is an index into the inputs of the system and specifies which input to use for the frequency response.

The command

$$\text{nyquist}(A,B,C,D,iu,w)$$

uses the user-supplied frequency vector w . The vector w specifies the frequencies in radians per second at which the frequency response should be calculated.

EXAMPLE 8–11 Consider the system defined by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -25 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 25 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

Draw a Nyquist plot.

This system has a single input u and a single output y . A Nyquist plot may be obtained by entering the command

$$\text{nyquist}(A,B,C,D)$$

or

$$\text{nyquist}(A,B,C,D,1)$$

MATLAB Program 8–12 will provide the Nyquist plot. (Note that we obtain the identical result by using either of these two commands.) Figure 8–45 shows the Nyquist plot produced by MATLAB Program 8–12.

MATLAB Program 8–12
<pre> A = [0 1;-25 -4]; B = [0;25]; C = [1 0]; D = [0]; nyquist(A,B,C,D) grid title('Nyquist Plot') </pre>

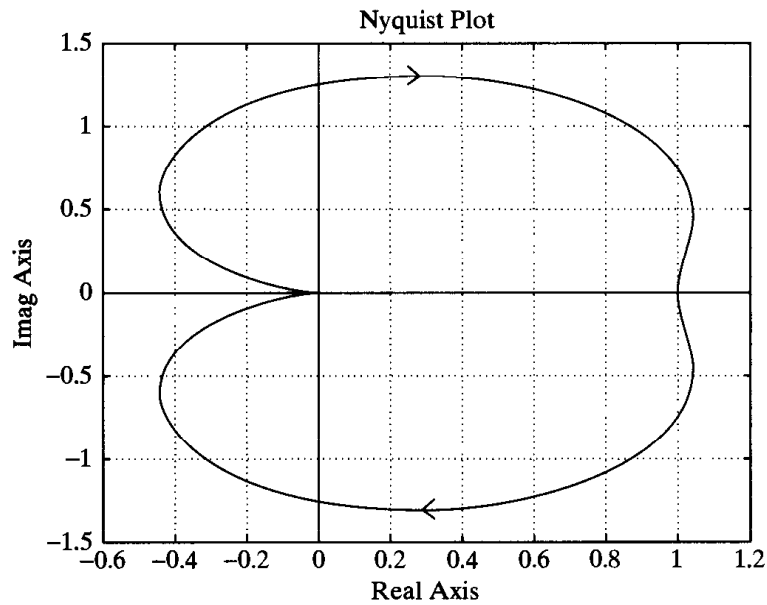


Figure 8–45
Nyquist plot of sys-
tem considered in
Example 8–11.

EXAMPLE 8–12 Consider the system defined by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 6.5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

This system involves two inputs and two outputs. There are four sinusoidal output–input relationships: $Y_1(j\omega)/U_1(j\omega)$, $Y_2(j\omega)/U_1(j\omega)$, $Y_1(j\omega)/U_2(j\omega)$, and $Y_2(j\omega)/U_2(j\omega)$. Draw Nyquist plots for the system. (When considering input u_1 , we assume that input u_2 is zero, and vice versa.)

The four individual Nyquist plots can be obtained by use of the command

nyquist(A,B,C,D)

MATLAB Program 8–13 produces the four Nyquist plots. They are shown in Figure 8–46.

MATLAB Program 8–13
$A = [-1 \quad -1; 6.5 \quad 0];$
$B = [1 \quad 1; 1 \quad 0];$
$C = [1 \quad 0; 0 \quad 1];$
$D = [0 \quad 0; 0 \quad 0];$
nyquist(A,B,C,D)

8–6 LOG-MAGNITUDE VERSUS PHASE PLOTS

Another approach to graphically portraying the frequency-response characteristics is to use the log-magnitude versus phase plot, which is a plot of the logarithmic magnitude in decibels versus the phase angle or phase margin for a frequency range of interest. [The phase margin is the difference between the actual phase angle ϕ and -180° ; that

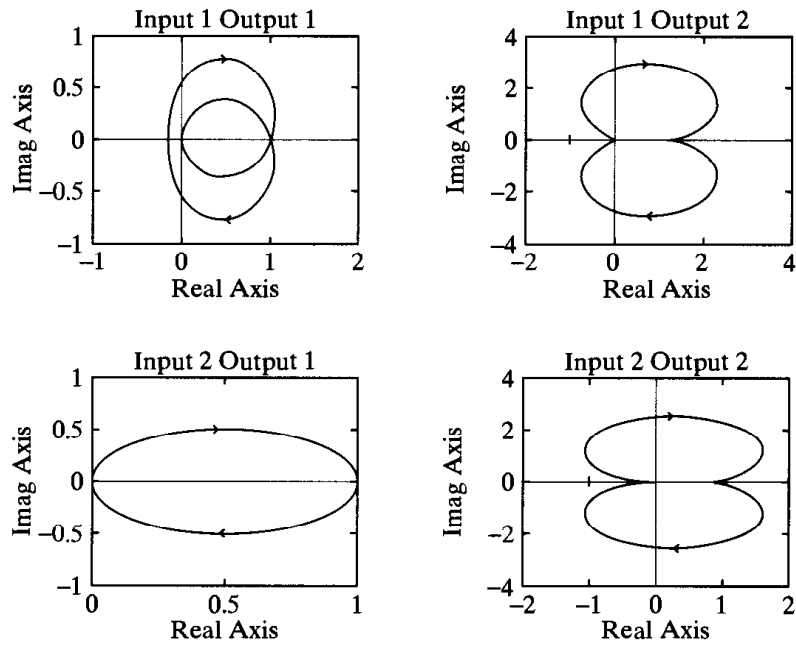


Figure 8-46
Nyquist plots of sys-
tem considered in
Example 8-12.

is, $\phi - (-180^\circ) = 180^\circ + \phi$.] The curve is graduated in terms of the frequency ω . Such log-magnitude versus phase plots are commonly called Nichols plots.

In the Bode diagram, the frequency-response characteristics of $G(j\omega)$ are shown on semilog paper by two separate curves, the log-magnitude curve and the phase-angle curve, while in the log-magnitude versus phase plot, the two curves in the Bode diagram are combined into one. The log-magnitude versus phase plot can easily be constructed by reading values of the log magnitude and phase angle from the Bode diagram. Notice that in the log-magnitude versus phase plot a change in the gain constant of $G(j\omega)$ merely shifts the curve up (for increasing gain) or down (for decreasing gain), but the shape of the curve remains the same.

Advantages of the log-magnitude versus phase plot are that the relative stability of the closed-loop system can be determined quickly and that compensation can be worked out easily.

The log-magnitude versus phase plots for the sinusoidal transfer function $G(j\omega)$ and $1/G(j\omega)$ are skew symmetrical about the origin since

$$\left| \frac{1}{G(j\omega)} \right| \text{ in dB} = -|G(j\omega)| \text{ in dB}$$

and

$$\angle \frac{1}{G(j\omega)} = -\angle G(j\omega)$$

Figure 8-47 compares frequency-response curves of

$$G(j\omega) = \frac{1}{1 + 2\xi \left(j \frac{\omega}{\omega_n} \right) + \left(j \frac{\omega}{\omega_n} \right)^2}$$

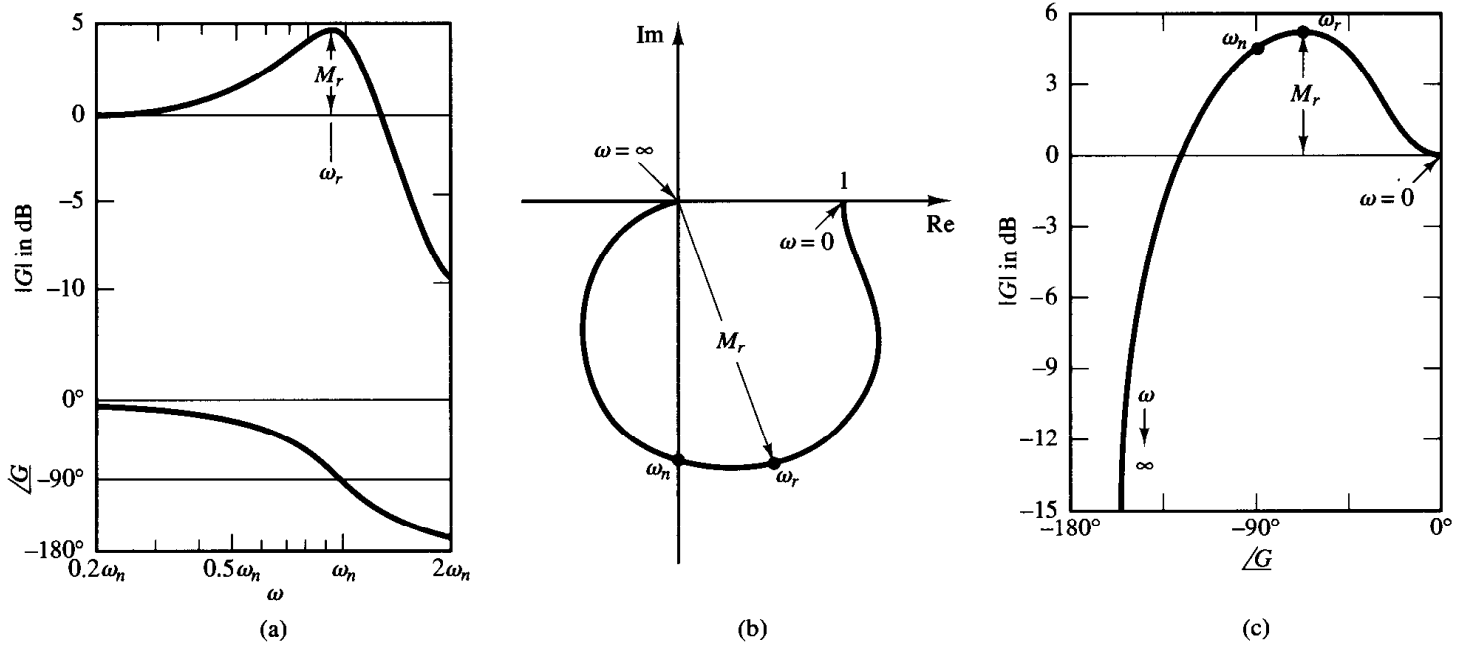


Figure 8-47

Three representations of the frequency response of $\frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2}$, for $\zeta > 0$.

(a) Bode diagram; (b) polar plot; (c) log-magnitude versus phase plot.

in three different representations. In the log-magnitude versus phase plot, the vertical distance between the points $\omega = 0$ and $\omega = \omega_r$, where ω_r is the resonant frequency, is the peak value of $G(j\omega)$ in decibels.

Since log-magnitude and phase-angle characteristics of basic transfer functions have been discussed in detail in Sections 8-2 and 8-3, it will be sufficient here to give examples of some log-magnitude versus phase plots. Table 8-2 shows such examples.

8-7 NYQUIST STABILITY CRITERION

This section presents the Nyquist stability criterion and associated mathematical backgrounds. Consider the closed-loop system shown in Figure 8-48. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

For stability, all roots of the characteristic equation

$$1 + G(s)H(s) = 0$$

Table 8–2 Log-Magnitude versus Phase Plots of Simple Transfer Functions

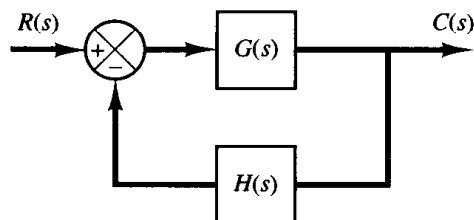
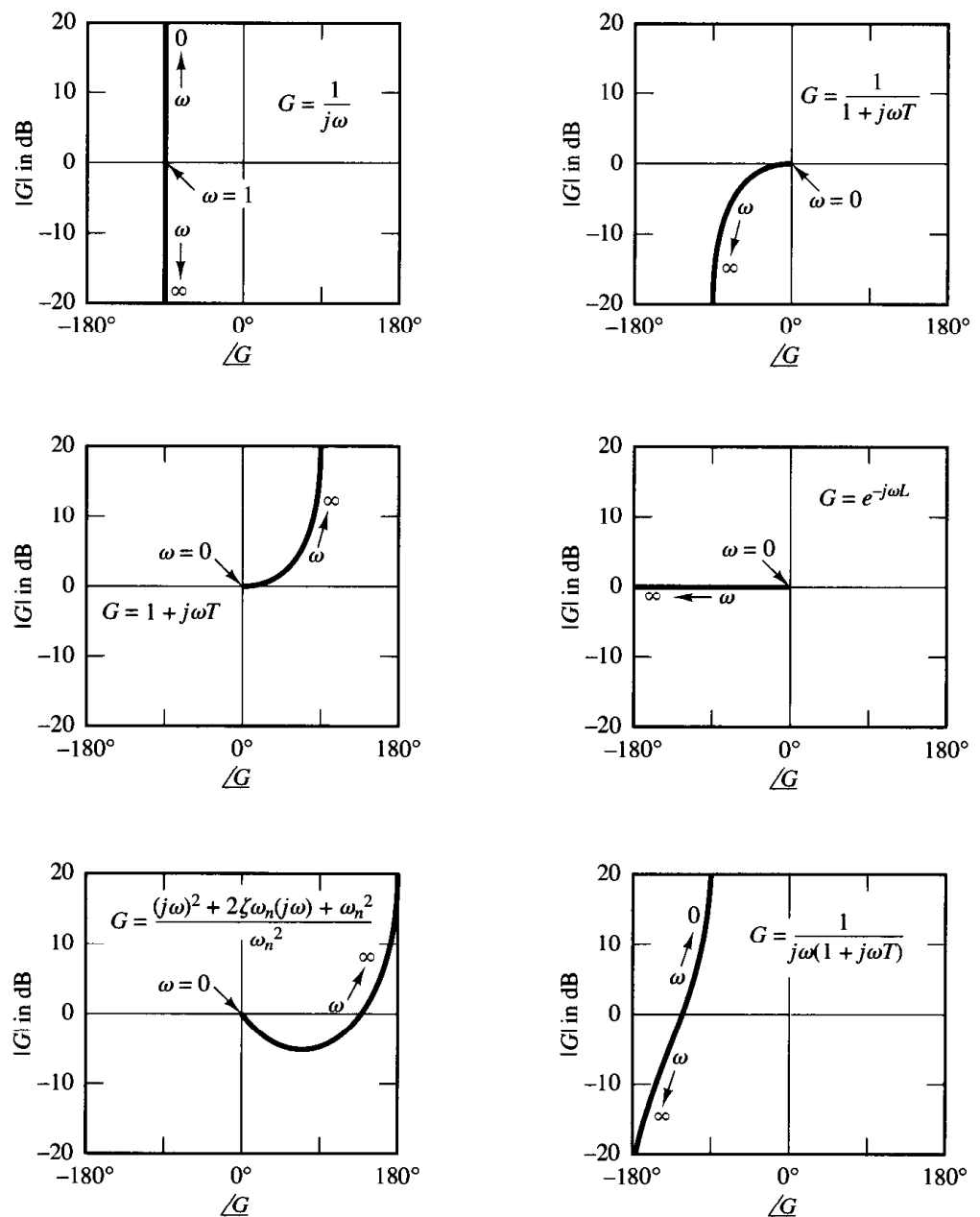


Figure 8–48
Closed-loop system.

must lie in the left-half s plane. [It is noted that, although poles and zeros of the open-loop transfer function $G(s)H(s)$ may be in the right-half s plane, the system is stable if all the poles of the closed-loop transfer function (that is, the roots of the characteristic equation) are in the left-half s plane.] The Nyquist stability criterion relates the open-loop frequency response $G(j\omega)H(j\omega)$ to the number of zeros and poles of $1 + G(s)H(s)$ that lie in the right-half s plane. This criterion, derived by H. Nyquist, is useful in control engineering because the absolute stability of the closed-loop system can be determined graphically from open-loop frequency-response curves, and there is no need for actually determining the closed-loop poles. Analytically obtained open-loop frequency-response curves, as well as those experimentally obtained, can be used for the stability analysis. This is convenient because, in designing a control system, it often happens that mathematical expressions for some of the components are not known; only their frequency-response data are available.

The Nyquist stability criterion is based on a theorem from the theory of complex variables. To understand the criterion, we shall first discuss mappings of contours in the complex plane.

We shall assume that the open-loop transfer function $G(s)H(s)$ is representable as a ratio of polynomials in s . For a physically realizable system, the degree of the denominator polynomial of the closed-loop transfer function must be greater than or equal to that of the numerator polynomial. This means that the limit of $G(s)H(s)$ as s approaches infinity is zero or a constant for any physically realizable system.

Preliminary study. The characteristic equation of the system shown in Figure 8–48 is

$$F(s) = 1 + G(s)H(s) = 0$$

We shall show that for a given continuous closed path in the s plane, which does not go through any singular points, there corresponds a closed curve in the $F(s)$ plane. The number and direction of encirclements of the origin of the $F(s)$ plane by the closed curve plays a particularly important role in what follows, for later we shall correlate the number and direction of encirclements with the stability of the system.

Consider, for example, the following open-loop transfer function:

$$G(s)H(s) = \frac{6}{(s + 1)(s + 2)}$$

The characteristic equation is

$$\begin{aligned} F(s) = 1 + G(s)H(s) &= 1 + \frac{6}{(s + 1)(s + 2)} \\ &= \frac{(s + 1.5 + j2.4)(s + 1.5 - j2.4)}{(s + 1)(s + 2)} = 0 \end{aligned}$$

The function $F(s)$ is analytic everywhere in the s plane except at its singular points. For each point of analyticity in the s plane, there corresponds a point in the $F(s)$ plane. For example, if $s = 1 + j2$, then $F(s)$ becomes

$$F(1 + j2) = 1 + \frac{6}{(2 + j2)(3 + j2)} = 1.115 - j0.577$$

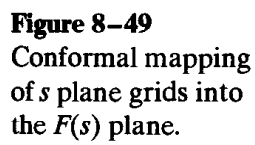
Thus, the points $s = 1 + j2$ in the s plane maps into the point $1.115 - j0.577$ in the $F(s)$ plane.

Thus, as stated previously, for a given continuous closed path in the s plane, which does not go through any singular points, there corresponds a closed curve in the $F(s)$ plane. Figure 8-49 (a) shows conformal mappings of the lines $\omega = 0, 1, 2, 3$ and the lines $\sigma = 1, 0, -1, -2, -3, -4$ in the upper-half s plane into the $F(s)$ plane. For example, the line $s = j\omega$ in the upper-half s plane ($\omega \geq 0$) maps into the curve denoted by $\sigma = 0$ in the $F(s)$ plane. Figure 8-49(b) shows conformal mappings of the lines $\omega = 0, -1, -2, -3$ and the lines $\sigma = 1, 0, -1, -2, -3, -4$ in the lower-half s plane into the $F(s)$ plane. Notice that for a given σ the curve for negative frequencies is symmetrical about the real axis with the curve for positive frequencies. Referring to Figures 8-49(a) and (b), we see that for the path $ABCD$ in the s plane traversed in the clockwise direction the corresponding curve in the $F(s)$ plane is $A'B'C'D'$. The arrows on the curves indicate directions of traversal. Similarly, the path $DEFA$ in the s plane maps into the curve $D'E'F'A'$ in the $F(s)$ plane. Because of the property of conformal mapping, the corresponding angles in the s plane and $F(s)$ plane are equal and have the same sense. [For example, since lines AB and BC intersect at right angles to each other in the s plane, curves $A'B'$ and $B'C'$ also intersect at right angles at point B' in the $F(s)$ plane.] Referring to Figure 8-49(c), we see that on the closed contour $ABCDEFA$ in the s plane the variable s starts at point A and assumes values on this path in a clockwise direction until it returns to the starting point A . The corresponding curve in the $F(s)$ plane is denoted $A'B'C'D'E'F'A'$. If we define the area to the right of the contour when a representative point s moves in the clockwise direction to be the inside of the contour and the area to the left to be the outside, then the shaded area in Figure 8-49(c) is enclosed by the contour $ABCDEFA$ and is inside it. From Figure 8-49(c), it can be seen that when the contour in the s plane encloses two poles of $F(s)$ the locus of $F(s)$ encircles the origin of the $F(s)$ plane twice in the counterclockwise direction.

The number of encirclements of the origin of the $F(s)$ plane depends on the closed contour in the s plane. If this contour encloses two zeros and two poles of $F(s)$, then the corresponding $F(s)$ locus does not encircle the origin, as shown in Figure 8-49(d). If this contour encloses only one zero, the corresponding locus of $F(s)$ encircles the origin once in the clockwise direction. This is shown in Figure 8-49(e). Finally, if the closed contour in the s plane encloses neither zeros nor poles, then the locus of $F(s)$ does not encircle the origin of the $F(s)$ plane at all. This is also shown in Figure 8-49(e).

Note that for each point in the s plane, except for the singular points, there is only one corresponding point in the $F(s)$ plane; that is, the mapping from the s plane into the $F(s)$ plane is one to one. The mapping from the $F(s)$ plane into the s plane may not be one to one, however, so that a given point in the $F(s)$ plane may correspond to more than one point in the s plane. For example, point B' in the $F(s)$ plane in Figure 8-49(d) corresponds to both point $(-3, 3)$ and point $(0, -3)$ in the s plane.

From the foregoing analysis, we can see that the direction of encirclement of the origin of the $F(s)$ plane depends on whether the contour in the s plane encloses a pole or a zero. Note that the location of a pole or zero in the s plane, whether in the right-half or left-half s plane, does not make any difference, but the enclosure of a pole or zero does. If the contour in the s plane encloses k zeros and k poles ($k = 0, 1, 2, \dots$), that is, an equal number of each, then the corresponding closed curve in the $F(s)$ plane does not encircle the origin of the $F(s)$ plane. The foregoing discussion is a



graphical explanation of the mapping theorem, which is the basis for the Nyquist stability criterion.

Mapping theorem. Let $F(s)$ be a ratio of two polynomials in s . Let P be the number of poles and Z be the number of zeros of $F(s)$ that lie inside some closed contour in the s plane, with multiplicity of poles and zeros accounted for. Let this contour be such that it does not pass through any poles or zeros of $F(s)$. This closed contour in the s plane is then mapped into the $F(s)$ plane as a closed curve. The total number N of clockwise encirclements of the origin of the $F(s)$ plane, as a representative point s traces out the entire contour in the clockwise direction, is equal to $Z - P$. (Note that by this mapping theorem the numbers of zeros and of poles cannot be found, only their difference.)

We shall not present a formal proof of this theorem here but leave the proof to Problem A-8-10. Note that a positive number N indicates an excess of zeros over poles of the function $F(s)$ and a negative N indicates an excess of poles over zeros. In control system applications, the number P can be readily determined for $F(s) = 1 + G(s)H(s)$ from the function $G(s)H(s)$. Therefore, if N is determined from the plot of $F(s)$, the number of zeros in the closed contour in the s plane can be determined readily. Note that the exact shapes of the s -plane contour and $F(s)$ locus are immaterial so far as encirclements of the origin are concerned, since encirclements depend only on the enclosure of poles and/or zeros of $F(s)$ by the s -plane contour.

Application of the mapping theorem to the stability analysis of closed-loop systems. For analyzing the stability of linear control systems, we let the closed contour in the s plane enclose the entire right-half s plane. The contour consists of the entire $j\omega$ axis from $\omega = -\infty$ to $+\infty$ and a semicircular path of infinite radius in the right-half s plane. Such a contour is called the Nyquist path. (The direction of the path is clockwise.) The Nyquist path encloses the entire right-half s plane and encloses all the zeros and poles of $1 + G(s)H(s)$ that have positive real parts. [If there are no zeros of $1 + G(s)H(s)$ in the right-half s plane, then there are no closed-loop poles there, and the system is stable.] It is necessary that the closed contour, or the Nyquist path, not pass through any zeros and poles of $1 + G(s)H(s)$. If $G(s)H(s)$ has a pole or poles at the origin of the s plane, mapping of the point $s = 0$ becomes indeterminate. In such cases, the origin is avoided by taking a detour around it. (A detailed discussion of this special case is given later.)

If the mapping theorem is applied to the special case in which $F(s)$ is equal to $1 + G(s)H(s)$, then we can make the following statement: If the closed contour in the s plane encloses the entire right-half s plane, as shown in Figure 8-50, then the number of right-half plane zeros of the function $F(s) = 1 + G(s)H(s)$ is equal to the number of poles of the function $F(s) = 1 + G(s)H(s)$ in the right-half s plane plus the number of clockwise encirclements of the origin of the $1 + G(s)H(s)$ plane by the corresponding closed curve in this latter plane.

Because of the assumed condition that

$$\lim_{s \rightarrow \infty} [1 + G(s)H(s)] = \text{constant}$$

the function of $1 + G(s)H(s)$ remains constant as s traverses the semicircle of infinite radius. Because of this, whether the locus of $1 + G(s)H(s)$ encircles the origin of the

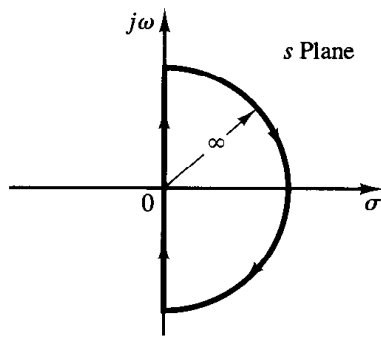


Figure 8-50
Closed contour in the s plane.

$1 + G(s)H(s)$ plane can be determined by considering only a part of the closed contour in the s plane, that is, the $j\omega$ axis. Encirclements of the origin, if there are any, occur only while a representative point moves from $-\infty$ to $+\infty$ along the $j\omega$ axis, provided that no zeros or poles lie on the $j\omega$ axis.

Note that the portion of the $1 + G(s)H(s)$ contour from $\omega = -\infty$ to $\omega = \infty$ is simply $1 + G(j\omega)H(j\omega)$. Since $1 + G(j\omega)H(j\omega)$ is the vector sum of the unit vector and the vector $G(j\omega)H(j\omega)$, $1 + G(j\omega)H(j\omega)$ is identical to the vector drawn from the $-1 + j0$ point to the terminal point of the vector $G(j\omega)H(j\omega)$, as shown in Figure 8-51. Encirclement of the origin by the graph of $1 + G(j\omega)H(j\omega)$ is equivalent to encirclement of the $-1 + j0$ point by just the $G(j\omega)H(j\omega)$ locus. Thus, stability of a closed-loop system can be investigated by examining encirclements of the $-1 + j0$ point by the locus of $G(j\omega)H(j\omega)$. The number of clockwise encirclements of the $-1 + j0$ point can be found by drawing a vector from the $-1 + j0$ point to the $G(j\omega)H(j\omega)$ locus, starting from $\omega = -\infty$, going through $\omega = 0$, and ending at $\omega = +\infty$, and by counting the number of clockwise rotations of the vector.

Plotting $G(j\omega)H(j\omega)$ for the Nyquist path is straightforward. The map of the negative $j\omega$ axis is the mirror image about the real axis of the map of the positive $j\omega$ axis. That is, the plot of $G(j\omega)H(j\omega)$ and the plot of $G(-j\omega)H(-j\omega)$ are symmetrical with each other about the real axis. The semicircle with infinite radius maps into either the origin of the GH plane or a point on the real axis of the GH plane.

In the preceding discussion, $G(s)H(s)$ has been assumed to be the ratio of two polynomials in s . Thus, the transport lag e^{-Ts} has been excluded from the discussion. Note, however, that a similar discussion applies to systems with transport lag, although a proof of this is not given here. The stability of a system with transport lag can be determined

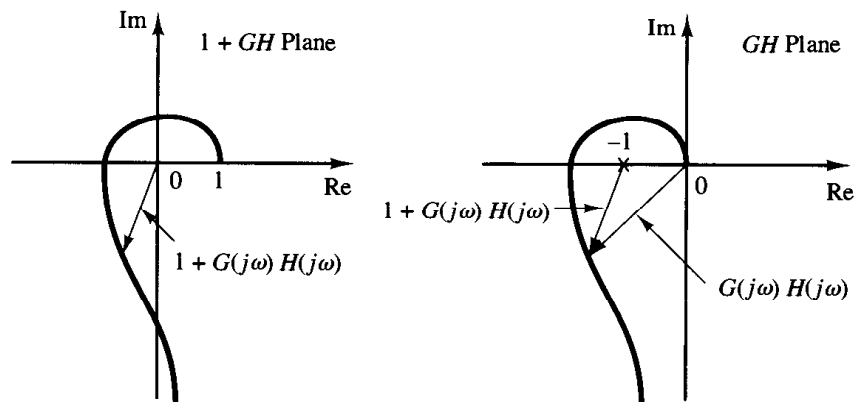


Figure 8-51
Plots of
 $1 + G(j\omega)H(j\omega)$ in
the $1 + GH$ plane
and GH plane.

from the open-loop frequency-response curves by examining the number of encirclements of the $-1 + j0$ point, just as in the case of a system whose open-loop transfer function is a ratio of two polynomials in s .

Nyquist stability criterion. The foregoing analysis, utilizing the encirclement of the $-1 + j0$ point by the $G(j\omega)H(j\omega)$ locus, is summarized in the following Nyquist stability criterion:

Nyquist stability criterion [for a special case when $G(s)H(s)$ has neither poles nor zeros on the $j\omega$ axis.]: In the system shown in Figure 8–48, if the open-loop transfer function $G(s)H(s)$ has k poles in the right-half s plane and $\lim_{s \rightarrow \infty} G(s)H(s) = \text{constant}$, then for stability the $G(j\omega)H(j\omega)$ locus, as ω varies from $-\infty$ to ∞ , must encircle the $-1 + j0$ point k times in the counterclockwise direction.

Remarks on the Nyquist stability criterion

1. This criterion can be expressed as

$$Z = N + P$$

where Z = number of zeros of $1 + G(s)H(s)$ in the right-half s plane

N = number of clockwise encirclements of the $-1 + j0$ point

P = number of poles of $G(s)H(s)$ in the right-half s plane

If P is not zero, for a stable control system, we must have $Z = 0$, or $N = -P$, which means that we must have P counterclockwise encirclements of the $-1 + j0$ point.

If $G(s)H(s)$ does not have any poles in the right-half s plane, then $Z = N$. Thus, for stability there must be no encirclement of the $-1 + j0$ point by the $G(j\omega)H(j\omega)$ locus. In this case it is not necessary to consider the locus for the entire $j\omega$ axis, only for the positive-frequency portion. The stability of such a system can be determined by seeing if the $-1 + j0$ point is enclosed by the Nyquist plot of $G(j\omega)H(j\omega)$. The region enclosed by the Nyquist plot is shown in Figure 8–52. For stability, the $-1 + j0$ point must lie outside the shaded region.

2. We must be careful when testing the stability of multiple-loop systems since they may include poles in the right-half s plane. (Note that although an inner loop may be unstable the entire closed-loop system can be made stable by proper design.) Simple in-

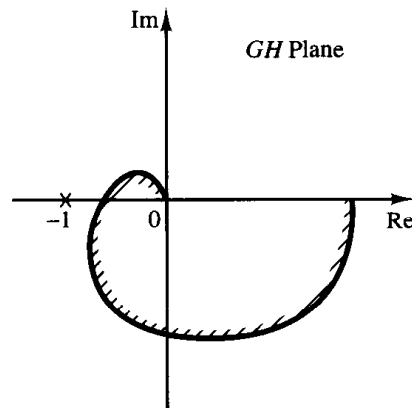


Figure 8–52
Region enclosed by a Nyquist plot.

spection of encirclements of the $-1 + j0$ point by the $G(j\omega)H(j\omega)$ locus is not sufficient to detect instability in multiple-loop systems. In such cases, however, whether any pole of $1 + G(s)H(s)$ is in the right-half s plane can be determined easily by applying the Routh stability criterion to the denominator of $G(s)H(s)$.

If transcendental functions, such as transport lag e^{-Ts} , are included in $G(s)H(s)$, they must be approximated by a series expansion before the Routh stability criterion can be applied. One form of a series expansion of e^{-Ts} was given in Chapter 5 and repeated here:

$$e^{-Ts} = \frac{1 - \frac{Ts}{2} + \frac{(Ts)^2}{8} - \frac{(Ts)^3}{48} + \dots}{1 + \frac{Ts}{2} + \frac{(Ts)^2}{8} + \frac{(Ts)^3}{48} + \dots}$$

As a first approximation, we may take only the first two terms in the numerator and denominator, respectively, or

$$e^{-Ts} \doteq \frac{1 - \frac{Ts}{2}}{1 + \frac{Ts}{2}} = \frac{2 - Ts}{2 + Ts}$$

This gives a good approximation to transport lag for the frequency range $0 \leq \omega \leq (0.5/T)$. [Note that the magnitude of $(2 - j\omega T)/(2 + j\omega T)$ is always unity, and the phase lag of $(2 - j\omega T)/(2 + j\omega T)$ closely approximates that of transport lag within the stated frequency range.]

3. If the locus of $G(j\omega)H(j\omega)$ passes through the $-1 + j0$ point, then zeros of the characteristic equation, or closed-loop poles, are located on the $j\omega$ axis. This is not desirable for practical control systems. For a well-designed closed-loop system, none of the roots of the characteristic equation should lie on the $j\omega$ axis.

Special case when $G(s)H(s)$ involves poles and/or zeros on the $j\omega$ axis. In the previous discussion, we assumed that the open-loop transfer function $G(s)H(s)$ has neither poles nor zeros at the origin. We now consider the case where $G(s)H(s)$ involves poles and/or zeros on the $j\omega$ axis.

Since the Nyquist path must not pass through poles or zeros of $G(s)H(s)$, if the function $G(s)H(s)$ has poles or zeros at the origin (or on the $j\omega$ axis at points other than the origin), the contour in the s plane must be modified. The usual way of modifying the contour near the origin is to use a semicircle with the infinitesimal radius ϵ , as shown in Figure 8-53. A representative point s moves along the negative $j\omega$ axis from $-j\infty$ to $j0^-$. From $s = j0^-$ to $s = j0^+$, the point moves along the semicircle of radius ϵ (where $\epsilon \ll 1$) and then moves along the positive $j\omega$ axis from $j0^+$ to $j\infty$. From $s = j\infty$, the contour follows a semicircle with infinite radius, and the representative point moves back to the starting point. The area that the modified closed contour avoids is very small and approaches zero as the radius ϵ approaches zero. Therefore, all the poles and zeros, if any, in the right-half s plane are enclosed by this contour.

Consider, for example, a closed-loop system whose open-loop transfer function is given by

$$G(s)H(s) = \frac{K}{s(Ts + 1)}$$

The points corresponding to $s = j0+$ and $s = j0-$ on the locus of $G(s)H(s)$ in the $G(s)H(s)$ plane are $-j\infty$ and $j\infty$, respectively. On the semicircular path with radius ϵ (where $\epsilon \ll 1$), the complex variable s can be written

$$s = \epsilon e^{j\theta}$$

where θ varies from -90° to $+90^\circ$. Then $G(s)H(s)$ becomes

$$G(\epsilon e^{j\theta})H(\epsilon e^{j\theta}) = \frac{K}{\epsilon e^{j\theta}} = \frac{K}{\epsilon} e^{-j\theta}$$

The value K/ϵ approaches infinity as ϵ approaches zero, and $-\theta$ varies from 90° to -90° as a representative point s moves along the semicircle. Thus, the points $G(j0-)H(j0-) = j\infty$ and $G(j0+)H(j0+) = -j\infty$ are joined by a semicircle of infinite radius in the right-half GH plane. The infinitesimal semicircular detour around the origin maps into the GH plane as a semicircle of infinite radius. Figure 8-54 shows the s -plane contour and the $G(s)H(s)$ locus in the GH plane. Points A , B , and C on the s -plane contour map into the respective points A' , B' , and C' on the $G(s)H(s)$ locus. As seen from Figure 8-54, points D , E , and F on the semicircle of infinite radius in the s plane map into the origin of the GH plane. Since there is no pole in the right-half s plane and the $G(s)H(s)$ locus does not encircle the $-1 + j0$ point, there are no zeros of the function $1 + G(s)H(s)$ in the right-half s plane. Therefore, the system is stable.

For an open-loop transfer function $G(s)H(s)$ involving a $1/s^n$ factor (where $n = 2, 3, \dots$), the plot of $G(s)H(s)$ has n clockwise semicircles of infinite radius about the origin as a representative point s moves along the semicircle of radius ϵ (where $\epsilon \ll 1$). For example, consider the following open-loop transfer function:

$$G(s)H(s) = \frac{K}{s^2(Ts + 1)}$$

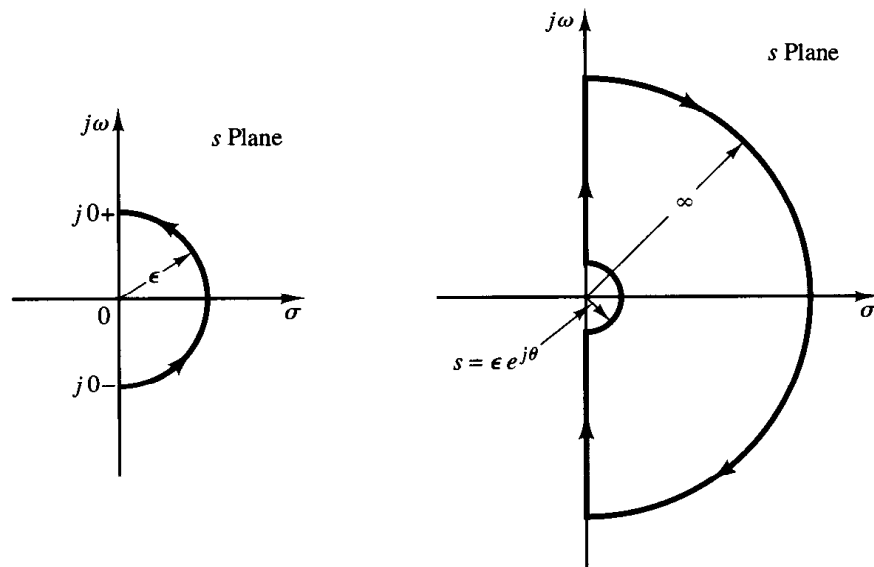


Figure 8-53
Closed contours in the s plane avoiding poles and zeros at the origin.

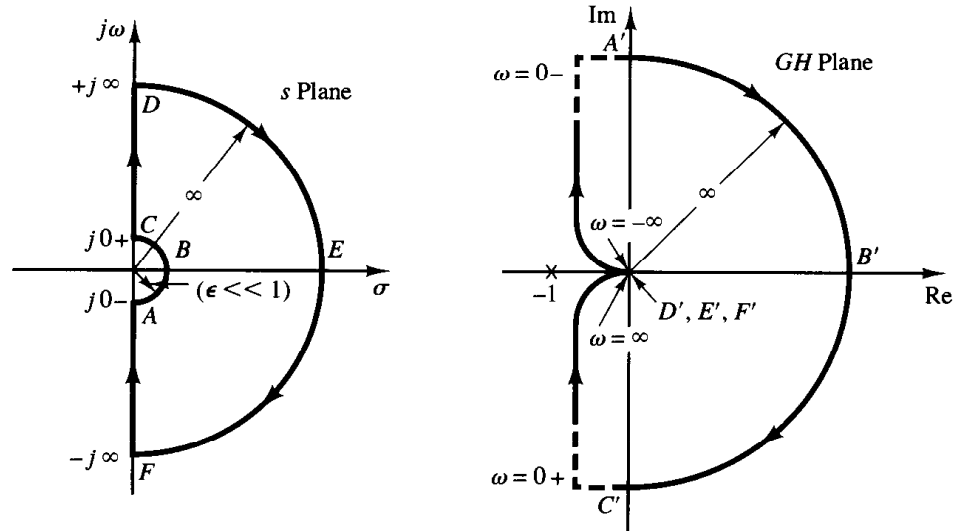


Figure 8-54
 s -Plane contour and
the $G(s)H(s)$ locus
in the GH plane,
where $G(s)H(s) =$
 $K/[s(Ts + 1)]$.

Then

$$\lim_{s \rightarrow \epsilon e^{j\theta}} G(s)H(s) = \frac{K}{\epsilon^2 e^{2j\theta}} = \frac{K}{\epsilon^2} e^{-2j\theta}$$

As θ varies from -90° to 90° in the s plane, the angle of $G(s)H(s)$ varies from 180° to -180° , as shown in Figure 8-55. Since there is no pole in the right-half s plane and the locus encircles the $-1 + j0$ point twice clockwise for any positive value of K , there are two zeros of $1 + G(s)H(s)$ in the right-half s plane. Therefore, this system is always unstable.

Note that a similar analysis can be made if $G(s)H(s)$ involves poles and/or zeros on the $j\omega$ axis. The Nyquist stability criterion can now be generalized as follows:

Nyquist stability criterion [for a general case when $G(s)H(s)$ has poles and/or zeros on the $j\omega$ axis.]: In the system shown in Figure 8-48, if the open-loop transfer function $G(s)H(s)$ has k poles in the right-half s plane, then for stability the $G(s)H(s)$ locus, as a representative point s traces on the modified Nyquist path in the

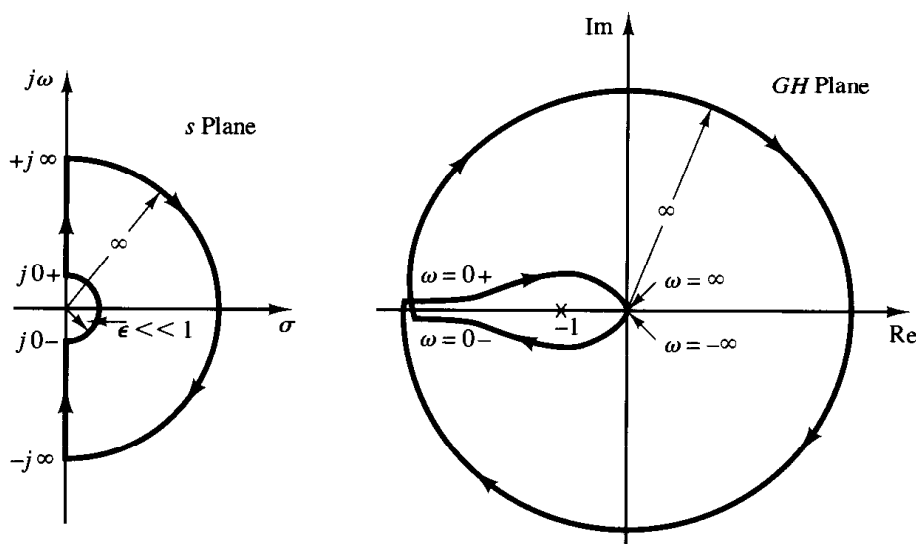


Figure 8-55
 s -Plane contour and
the $G(s)H(s)$ locus
in the GH plane,
where $G(s)H(s) =$
 $K/[s^2(Ts + 1)]$.

clockwise direction, must encircle the $-1 + j0$ point k times in the counterclockwise direction.

8-8 STABILITY ANALYSIS

In this section, we shall present several illustrative examples of the stability analysis of control systems using the Nyquist stability criterion.

If the Nyquist path in the s plane encircles Z zeros and P poles of $1 + G(s)H(s)$ and does not pass through any poles or zeros of $1 + G(s)H(s)$ as a representative point s moves in the clockwise direction along the Nyquist path, then the corresponding contour in the $G(s)H(s)$ plane encircles the $-1 + j0$ point $N = Z - P$ times in the clockwise direction. (Negative values of N imply counterclockwise encirclements.)

In examining the stability of linear control systems using the Nyquist stability criterion, we see that three possibilities can occur.

1. There is no encirclement of the $-1 + j0$ point. This implies that the system is stable if there are no poles of $G(s)H(s)$ in the right-half s plane; otherwise, the system is unstable.
2. There is a counterclockwise encirclement or encirclements of the $-1 + j0$ point. In this case the system is stable if the number of counterclockwise encirclements is the same as the number of poles $G(s)H(s)$ in the right-half s plane; otherwise, the system is unstable.
3. There is a clockwise encirclement or encirclements of the $-1 + j0$ point. In this case the system is unstable.

In the following examples, we assume that the values of the gain K and the time constants (such as T , T_1 , and T_2) are all positive.

EXAMPLE 8-13 Consider a closed-loop system whose open-loop transfer function is given by

$$G(s)H(s) = \frac{K}{(T_1s + 1)(T_2s + 1)}$$

Examine the stability of the system.

A plot of $G(j\omega)H(j\omega)$ is shown in Figure 8-56. Since $G(s)H(s)$ does not have any poles in the right-half s plane and the $-1 + j0$ point is not encircled by the $G(j\omega)H(j\omega)$ locus, this system is stable for any positive values of K , T_1 , and T_2 .

EXAMPLE 8-14 Consider the system with the following open-loop transfer function:

$$G(s) = \frac{K}{s(T_1s + 1)(T_2s + 1)}$$

Determine the stability of the system for two cases: (1) the gain K is small and (2) K is large.

The Nyquist plots of the open-loop transfer function with a small value of K and a large value of K are shown in Figure 8-57. The number of poles of $G(s)H(s)$ in the right-half s plane is zero. Therefore, for this system to be stable, it is necessary that $N = Z = 0$ or that the $G(s)H(s)$ locus not encircle the $-1 + j0$ point.

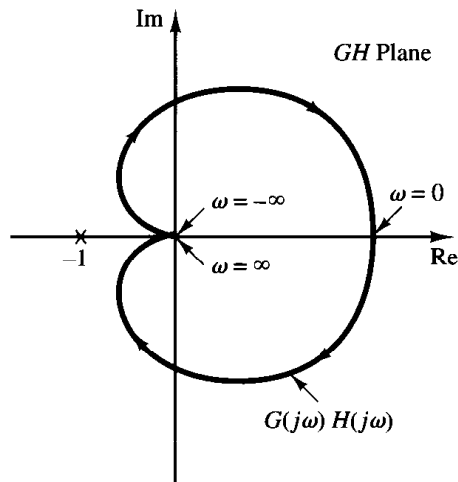


Figure 8-56
Polar plot of $G(j\omega)H(j\omega)$ considered in Example 8-13.

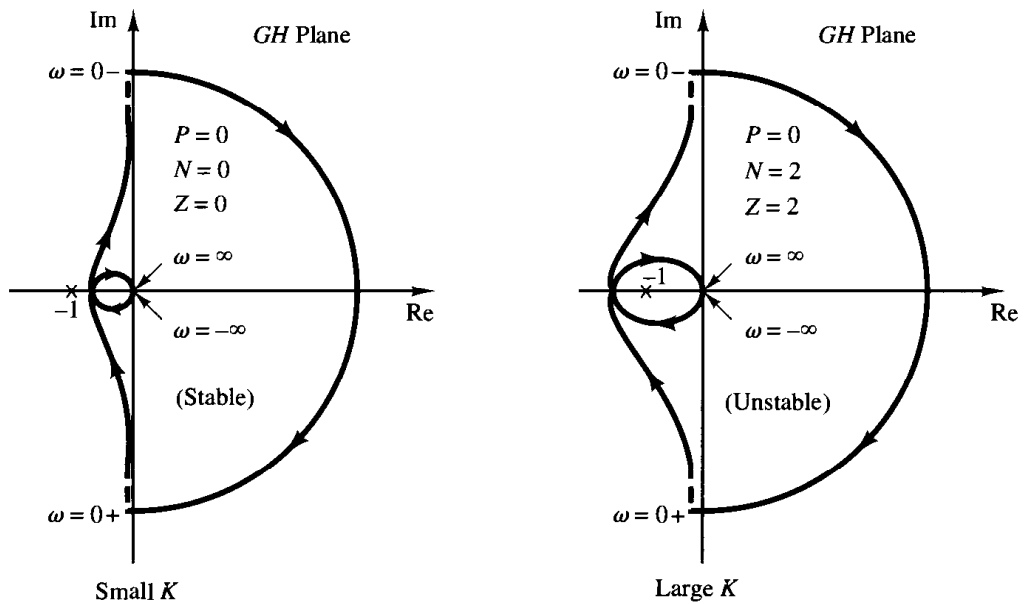


Figure 8-57
Polar plots of the system considered in Example 8-14.

For small values of K , there is no encirclement of the $-1 + j0$ point. Hence, the system is stable for small values of K . For large values of K , the locus of $G(s)H(s)$ encircles the $-1 + j0$ point twice in the clockwise direction, indicating two closed-loop poles in the right-half s plane, and the system is unstable. (For good accuracy, K should be large. From the stability viewpoint, however, a large value of K causes poor stability or even instability. To compromise between accuracy and stability, it is necessary to insert a compensation network into the system. Compensating techniques in the frequency domain are discussed in Chapter 9.)

EXAMPLE 8-15 The stability of a closed-loop system with the following open-loop transfer function

$$G(s)H(s) = \frac{K(T_2s + 1)}{s^2(T_1s + 1)}$$

depends on the relative magnitudes of T_1 and T_2 . Draw Nyquist plots and determine the stability of the system.

Plots of the locus $G(s)H(s)$ for three cases, $T_1 < T_2$, $T_1 = T_2$, and $T_1 > T_2$, are shown in Figure 8-58. For $T_1 < T_2$, the locus of $G(s)H(s)$ does not encircle the $-1 + j0$ point, and the closed-loop system is stable. For $T_1 = T_2$ the $G(s)H(s)$ locus passes through the $-1 + j0$ point, which

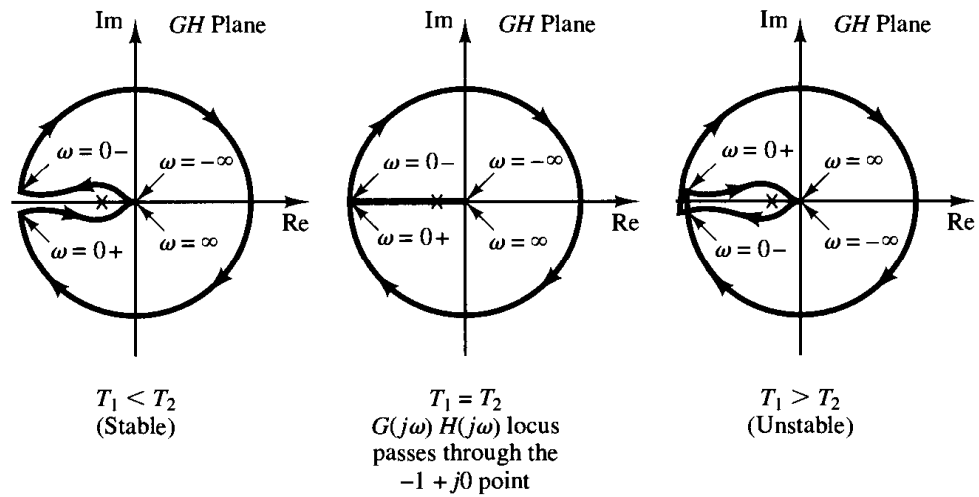


Figure 8-58
Polar plots of the system considered in Example 8-15.

indicates that the closed-loop poles are located on the $j\omega$ axis. For $T_1 > T_2$, the locus of $G(s)H(s)$ encircles the $-1 + j0$ point twice in the clockwise direction. Thus, the closed-loop system has two closed-loop poles in the right-half s plane, and the system is unstable.

EXAMPLE 8-16 Consider the closed-loop system having the following open-loop transfer function:

$$G(s)H(s) = \frac{K}{s(Ts - 1)}$$

Determine the stability of the system.

The function $G(s)H(s)$ has one pole ($s = 1/T$) in the right-half s plane. Therefore, $P = 1$. The Nyquist plot shown in Figure 8-59 indicates that the $G(s)H(s)$ plot encircles the $-1 + j0$ point once clockwise. Thus, $N = 1$. Since $Z = N + P$, we find that $Z = 2$. This means that the closed-loop system has two closed-loop poles in the right-half s plane and is unstable.

EXAMPLE 8-17 Investigate the stability of a closed-loop system with the following open-loop transfer function:

$$G(s)H(s) = \frac{K(s + 3)}{s(s - 1)} \quad (K > 1)$$

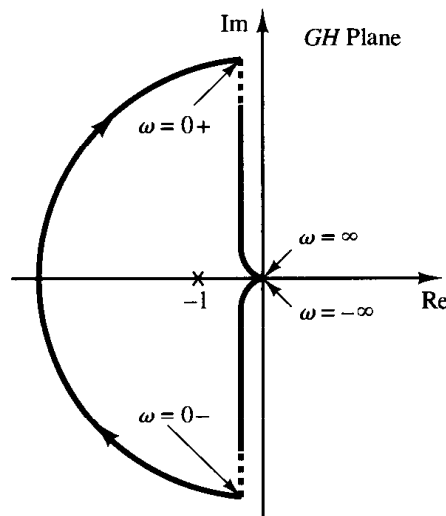


Figure 8-59
Polar plot of the system considered in Example 8-16.

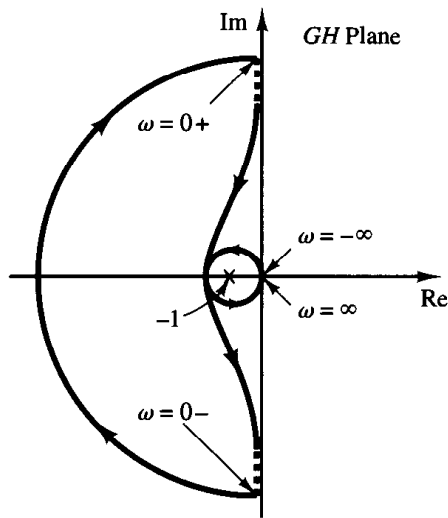


Figure 8-60
Polar plot of the system considered
in Example 8-17.

The open-loop transfer function has one pole ($s = 1$) in the right-half s plane, or $P = 1$. The open-loop system is unstable. The Nyquist plot shown in Figure 8-60 indicates that the $-1 + j0$ point is encircled by the $G(s)H(s)$ locus once in the counterclockwise direction. Therefore, $N = -1$. Thus, Z is found from $Z = N + P$ to be zero, which indicates that there is no zero of $1 + G(s)H(s)$ in the right-half s plane, and the closed-loop system is stable. This is one of the examples for which an unstable open-loop system becomes stable when the loop is closed.

Conditionally stable systems. Figure 8-61 shows an example of a $G(j\omega)H(j\omega)$ locus for which the closed-loop system can be made unstable by varying the open-loop gain. If the open-loop gain is increased sufficiently, the $G(j\omega)H(j\omega)$ locus encloses the $-1 + j0$ point twice, and the system becomes unstable. If the open-loop gain is decreased sufficiently, again the $G(j\omega)H(j\omega)$ locus encloses the $-1 + j0$ point twice. For stable operation of the system considered here, the critical point $-1 + j0$ must not be located in the regions between OA and BC shown in Figure 8-61. Such a system that is stable only for limited ranges of values of the open-loop gain for which the $-1 + j0$ point is completely outside the $G(j\omega)H(j\omega)$ locus is a conditionally stable system.

A conditionally stable system is stable for the value of the open-loop gain lying between critical values, but it is unstable if the open-loop gain is either increased or decreased sufficiently. Such a system becomes unstable when large input signals are

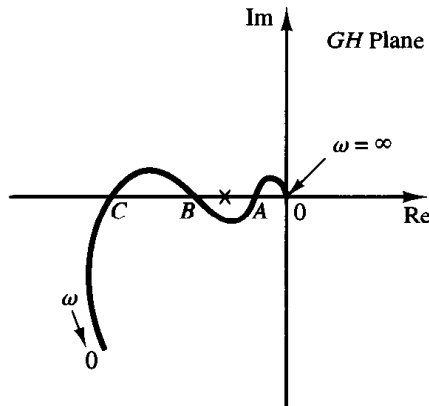


Figure 8-61
Polar plot of a conditionally
stable system.

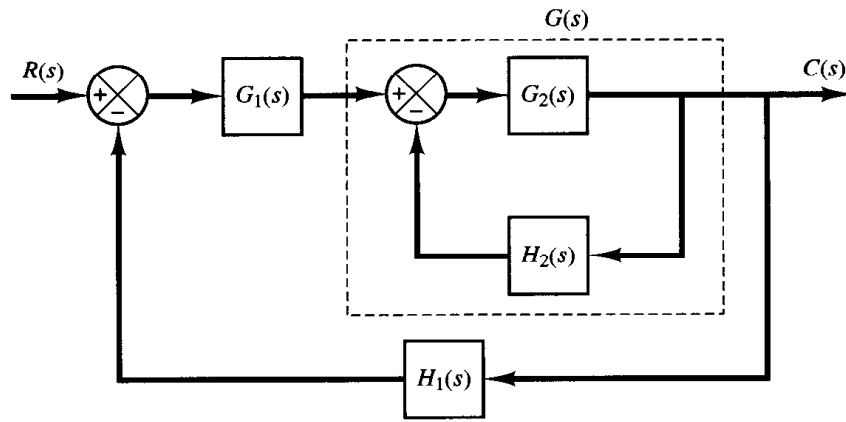


Figure 8-62
Multiple-loop system.

applied, since a large signal may cause saturation, which in turn reduces the open-loop gain of the system. It is advisable to avoid such a situation.

Multiple-loop system. Consider the system shown in Figure 8-62. This is a multiple-loop system. The inner loop has the transfer function

$$G(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s)}$$

If $G(s)$ is unstable, the effects of instability are to produce a pole or poles in the right-half s plane. Then the characteristic equation of the inner loop, $1 + G_2(s)H_2(s) = 0$, has a zero or zeros in this portion of the plane. If $G_2(s)$ and $H_2(s)$ have P_1 poles here, then the number Z_1 of right-half plane zeros of $1 + G_2(s)H_2(s)$ can be found from $Z_1 = N_1 + P_1$, where N_1 is the number of clockwise encirclements of the $-1 + j0$ point by the $G_2(s)H_2(s)$ locus. Since the open-loop transfer function of the entire system is given by $G_1(s)G(s)H_1(s)$, the stability of this closed-loop system can be found from the Nyquist plot of $G_1(s)G(s)H_1(s)$ and knowledge of the right-half plane poles of $G_1(s)G(s)H_1(s)$.

Notice that if a feedback loop is eliminated by means of block diagram reductions there is a possibility that unstable poles are introduced; if the feedforward branch is eliminated by means of block diagram reductions, there is a possibility that right-half plane zeros are introduced. Therefore, we must note all right-half plane poles and zeros as they appear from subsidiary loop reductions. This knowledge is necessary in determining the stability of multiple-loop systems.

EXAMPLE 8-18

Consider the control system shown in Figure 8-63. The system involves two loops. Determine the range of gain K for stability of the system by use of the Nyquist stability criterion. (The gain K is positive).

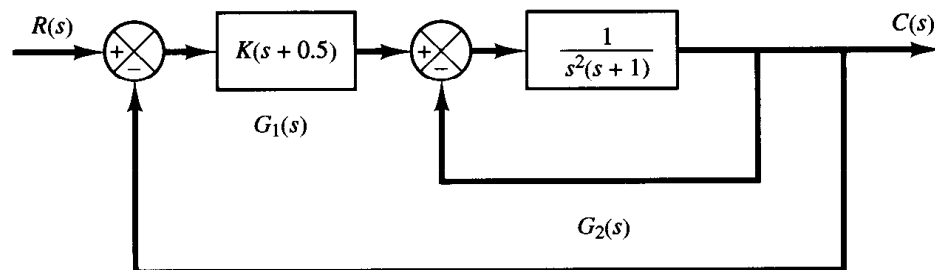


Figure 8-63
Control system.

To examine the stability of the control system, we need to sketch the Nyquist locus of $G(s)$, where

$$G(s) = G_1(s)G_2(s)$$

However, the poles of $G(s)$ are not known at this point. Therefore, we need to examine the minor loop if there are right-half s -plane poles. This can be done easily by use of the Routh stability criterion. Since

$$G_2(s) = \frac{1}{s^3 + s^2 + 1}$$

the Routh array becomes as follows:

$$\begin{array}{ccc} s^3 & 1 & 0 \\ s^2 & 1 & 1 \\ s^1 & -1 & 0 \\ s^0 & 1 & \end{array}$$

Notice that there are two sign changes in the first column. Hence, there are two poles of $G_2(s)$ in the right-half s plane.

Once we find the number of right-half s plane poles of $G_2(s)$, we proceed to sketch the Nyquist locus of $G(s)$, where

$$G(s) = G_1(s)G_2(s) = \frac{K(s + 0.5)}{s^3 + s^2 + 1}$$

Our problem is to determine the range of gain K for stability. Hence, instead of plotting Nyquist loci of $G(j\omega)$ for various values of K , we plot the Nyquist locus of $G(j\omega)/K$. Figure 8-64 shows the Nyquist plot or polar plot of $G(j\omega)/K$.

Since $G(s)$ has two poles in the right-half s plane, we have $P_1 = 2$. Noting that

$$Z_1 = N_1 + P_1$$

for stability, we require $Z_1 = 0$ or $N_1 = -2$. That is, the Nyquist locus of $G(j\omega)$ must encircle the $-1 + j0$ point twice counterclockwise. From Figure 8-64, we see that, if the critical point lies between 0 and -0.5 , then the $G(j\omega)/K$ locus encircles the critical point twice counterclockwise. Therefore, we require

$$-0.5K < -1$$

The range of gain K for stability is

$$2 < K$$

Nyquist stability criterion applied to inverse polar plots. In the previous analyses, the Nyquist stability criterion was applied to polar plots of the open-loop transfer function $G(s)H(s)$.

In analyzing multiple-loop systems, the inverse transfer function may sometimes be used in order to permit graphical analysis; this avoids much of the numerical calculation. (The Nyquist stability criterion can be applied equally well to inverse polar plots. The mathematical derivation of the Nyquist stability criterion for inverse polar plots is the same as that for direct polar plots.)

The inverse polar plot of $G(j\omega)H(j\omega)$ is a graph of $1/[G(j\omega)H(j\omega)]$ as a function of ω . For example, if $G(j\omega)H(j\omega)$ is

$$G(j\omega)H(j\omega) = \frac{j\omega T}{1 + j\omega T}$$

then

$$\frac{1}{G(j\omega)H(j\omega)} = \frac{1}{j\omega T} + 1$$

The inverse polar plot for $\omega \geq 0$ is the lower half of the vertical line starting at the point (1, 0) on the real axis.

The Nyquist stability criterion applied to inverse plots may be stated as follows: For a closed-loop system to be stable, the encirclement, if any, of the $-1 + j0$ point by the $1/[G(s)H(s)]$ locus (as s moves along the Nyquist path) must be counterclockwise, and the number of such encirclements must be equal to the number of poles of $1/[G(s)H(s)]$ [that is, the zeros of $G(s)H(s)$] that lie in the right-half s plane. [The number of zeros of $G(s)H(s)$ in the right-half s plane may be determined by use of the Routh stability criterion.] If the open-loop transfer function $G(s)H(s)$ has no zeros in the right-half s

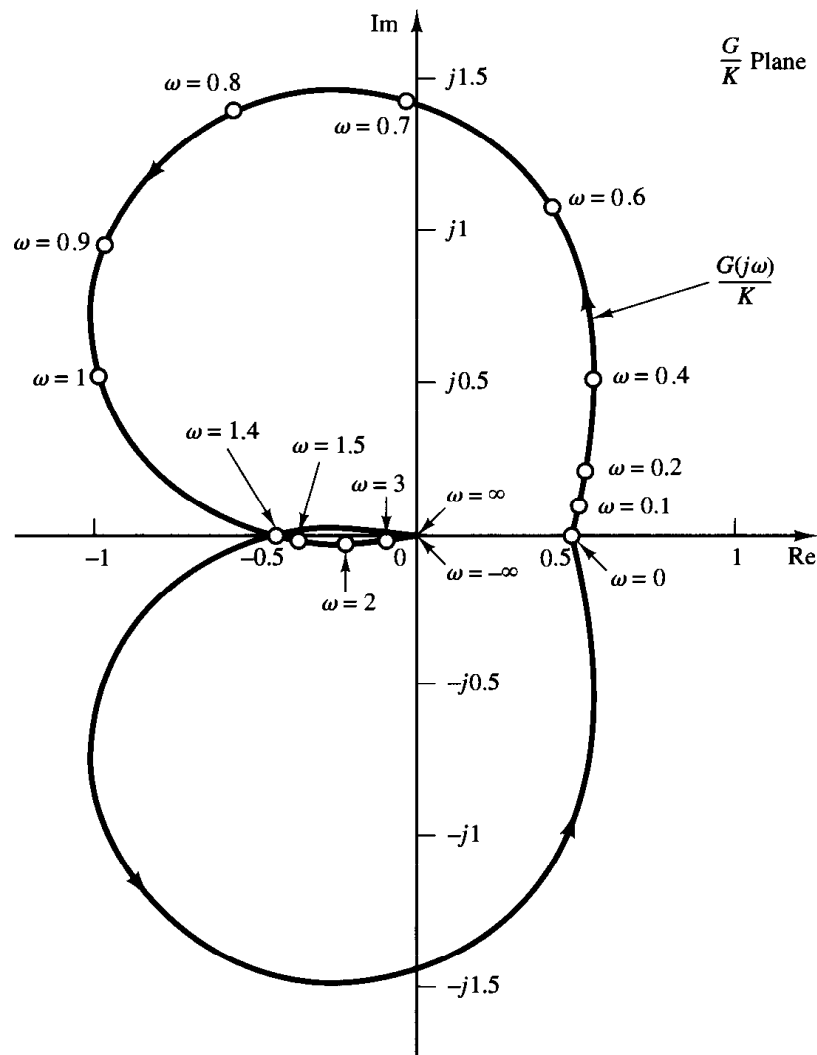


Figure 8-64
Polar plot of
 $G(j\omega)/K$.

plane, then for a closed-loop system to be stable the number of encirclements of the $-1 + j0$ point by the $1/[G(s)H(s)]$ locus must be zero.

Note that although the Nyquist stability criterion can be applied to inverse polar plots, if experimental frequency-response data are incorporated, counting the number of encirclements of the $1/[G(s)H(s)]$ locus may be difficult because the phase shift corresponding to the infinite semicircular path in the s plane is difficult to measure. For example, if the open-loop transfer function $G(s)H(s)$ involves transport lag such that

$$G(s)H(s) = \frac{Ke^{-j\omega L}}{s(Ts + 1)}$$

then the number of encirclements of the $-1 + j0$ point by the $1/[G(s)H(s)]$ locus becomes infinite, and the Nyquist stability criterion cannot be applied to the inverse polar plot of such an open-loop transfer function.

In general, if experimental frequency-response data cannot be put into analytical form, both the $G(j\omega)H(j\omega)$ and $1/[G(j\omega)H(j\omega)]$ loci must be plotted. In addition, the number of right-half plane zeros of $G(s)H(s)$ must be determined. It is more difficult to determine the right-half plane zeros of $G(s)H(s)$ (in other words, to determine whether a given component is minimum phase) than it is to determine the right-half plane poles of $G(s)H(s)$ (in other words, to determine whether the component is stable).

Depending on whether the data are graphical or analytical and whether nonminimum-phase components are included, an appropriate stability test must be used for multiple-loop systems. If the data are given in analytical form or if mathematical expressions for all the components are known, the application of the Nyquist stability criterion to inverse polar plots causes no difficulty, and multiple-loop systems may be analyzed and designed in the inverse GH plane.

EXAMPLE 8-19

Consider the control system shown in Figure 8-63. (Refer to Example 8-18.) Using the inverse polar plot, determine the range of gain K for stability.

Since

$$G_2(s) = \frac{1}{s^3 + s^2 + 1}$$

we have

$$G(s) = G_1(s)G_2(s) = \frac{K(s + 0.5)}{s^3 + s^2 + 1}$$

Hence

$$\frac{1}{G(s)} = \frac{s^3 + s^2 + 1}{K(s + 0.5)}$$

Notice that $1/G(s)$ has a pole at $s = -0.5$. It does not have any pole in the right-half s plane. Therefore, the Nyquist stability equation

$$Z = N + P$$

reduces to $Z = N$ since $P = 0$. The reduced equation states that the number Z of the zeros of $1 + [1/G(s)]$ in the right-half s plane is equal to N , the number of clockwise encirclements of the

$-1 + j0$ point. For stability, N must be equal to zero, or there should be no encirclement. Figure 8–65 shows the Nyquist plot or polar plot of $K/G(j\omega)$.

Notice that since

$$\begin{aligned}\frac{K}{G(j\omega)} &= \left[\frac{(j\omega)^3 + (j\omega)^2 + 1}{j\omega + 0.5} \right] \left(\frac{0.5 - j\omega}{0.5 - j\omega} \right) \\ &= \frac{0.5 - 0.5\omega^2 - \omega^4 + j\omega(-1 + 0.5\omega^2)}{0.25 + \omega^2}\end{aligned}$$

the $K/G(j\omega)$ locus crosses the negative real axis at $\omega = \sqrt{2}$, and the crossing point at the negative real axis is -2 .

From Figure 8–65 we see that if the critical point lies in the region between -2 and $-\infty$ then the critical point is not encircled. Hence, for stability we require

$$-1 < \frac{-2}{K}$$

Thus, the range of gain K for stability is

$$2 < K$$

which is the same result as we obtained in Example 8–18.

Relative stability analysis through modified Nyquist plots. The Nyquist path for stability tests can be modified in order that we may investigate the relative stability of closed-loop systems. For the following second-order characteristic equation,

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0 \quad (0 < \xi < 1)$$

the roots are complex conjugates and are

$$s_1 = -\xi\omega_n + j\omega_n\sqrt{1 - \xi^2}, \quad s_2 = -\xi\omega_n - j\omega_n\sqrt{1 - \xi^2}$$

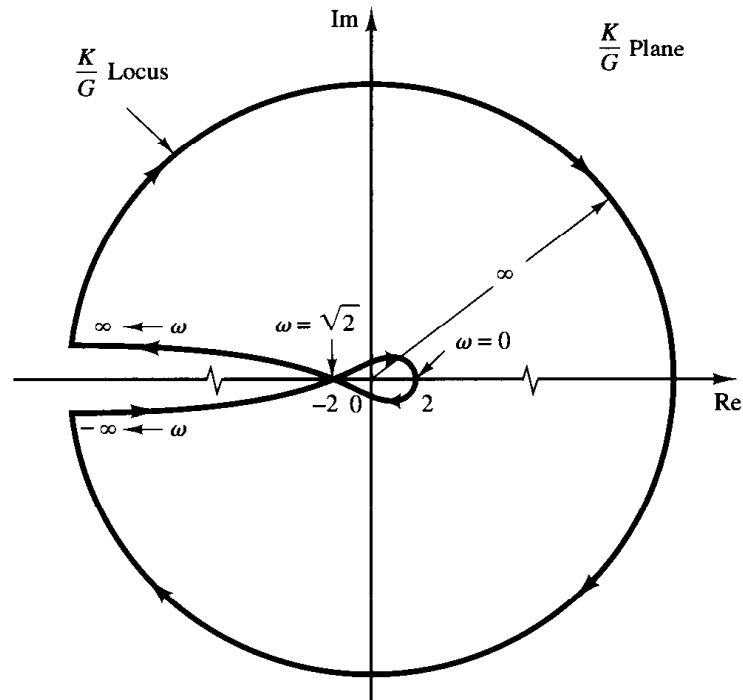


Figure 8–65
Polar plot of
 $K/G(j\omega)$.

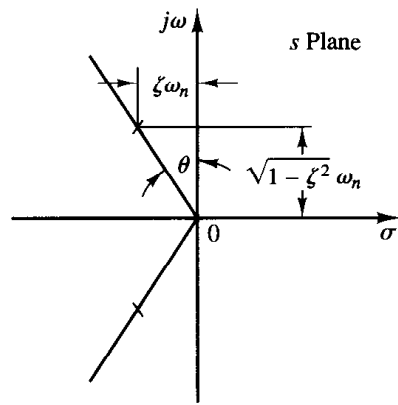


Figure 8-66
Plot of complex-conjugate roots in the s plane.

If these roots are plotted in the s plane, as shown in Figure 8-66, then we see that $\sin \theta = \zeta$, or the angle θ is indicative of the damping ratio ζ . As θ becomes smaller, so does the value of ζ .

If we modify the Nyquist path and use radial lines with angle θ_x , instead of the $j\omega$ axis, as shown in Figure 8-67, then it can be said, following the same reasoning as in the case of the Nyquist stability criterion, that if the $G(s)H(s)$ locus corresponding to the modified s -plane contour does not encircle the $-1 + j0$ point and none of the poles of $G(s)H(s)$ lie within the closed s -plane contour then this contour does not enclose any zeros of $1 + G(s)H(s)$. The characteristic equation, $1 + G(s)H(s) = 0$, then does not have any roots within the modified s -plane contour. If no closed-loop poles of a higher-order system are enclosed by this contour, we can say that the damping ratio of each pair of complex-conjugate closed-loop poles of the system is greater than $\sin \theta_x$.

Suppose that the s -plane contour consists of a line to the left of and parallel to the $j\omega$ axis at a distance $-\sigma_o$ (or the line $s = -\sigma_o + j\omega$) and the semicircle of infinite radius enclosing the entire right-half s plane and that part of the left-half s plane between the lines $s = -\sigma_o + j\omega$ and $s = j\omega$, as shown in Figure 8-68(a). If the $G(s)H(s)$ locus corresponding to this s -plane contour does not encircle the $-1 + j0$ point and $G(s)H(s)$ has no poles within the enclosed s -plane contour, then the characteristic equation does not have any zeros in the region enclosed by the modified s -plane contour. All roots of the characteristic equation lie to the left of the line $s = -\sigma_o + j\omega$. An example of a $G(-\sigma_o + j\omega)H(-\sigma_o + j\omega)$ locus, together with a $G(j\omega)H(j\omega)$ locus, is shown in Figure 8-68(b). The magnitude $1/\sigma_o$ is indicative of the time constant of the dominant closed-loop poles. If all

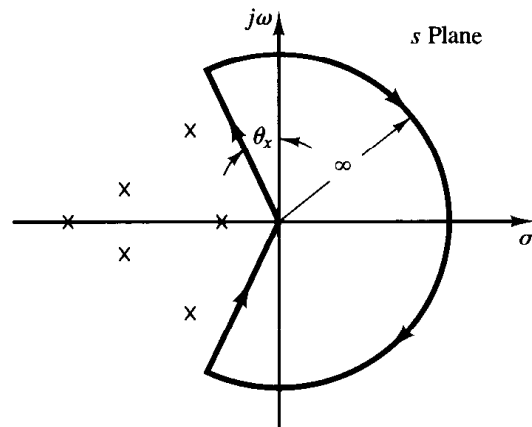


Figure 8-67
Modified Nyquist path.

Figure 8-68

(a) Modified Nyquist path; (b) polar plots of $G(-\sigma_o + j\omega)$ $H(-\sigma_o + j\omega)$ locus and $G(j\omega)H(j\omega)$ locus in the GH plane.

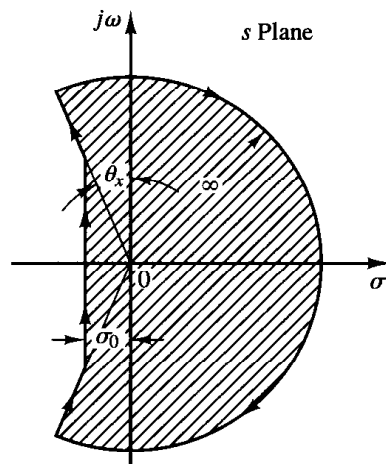
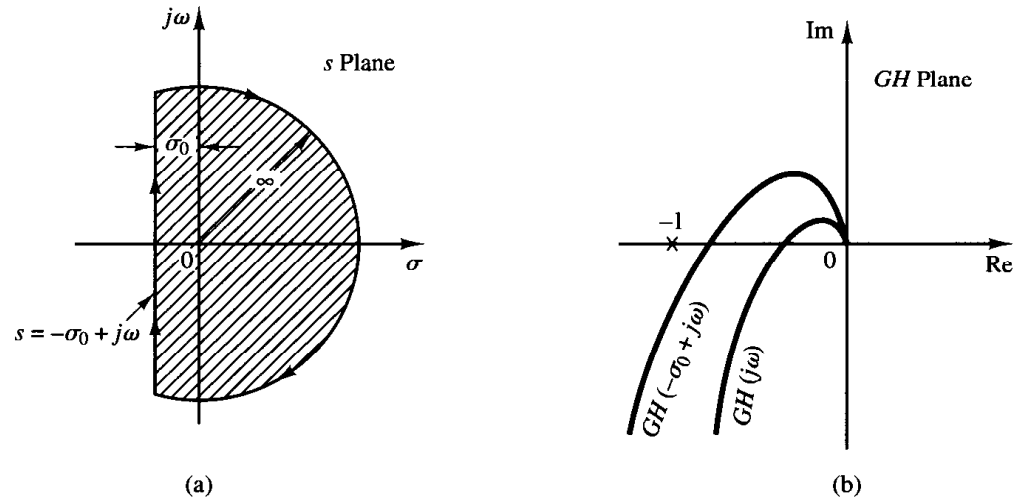


Figure 8-69

Modified Nyquist path.

roots lie outside the s -plane contour, all time constants of the closed-loop transfer function are less than $1/\sigma_o$. If the s -plane contour is chosen as shown in Figure 8-69, then the test of encirclements of the $-1 + j0$ point reveals the existence or nonexistence of the roots of the characteristic equation of the closed-loop system within this s -plane contour. If the test reveals that no roots lie in the s -plane contour, then it is clear that all the closed-loop poles have damping ratios greater than ζ_x and time constants less than $1/\sigma_o$. Thus, by taking an appropriate s -plane contour, we can investigate time constants and damping ratios of closed-loop poles from the open-loop transfer function.

8-9 RELATIVE STABILITY

In designing a control system, we require that the system be stable. Furthermore, it is necessary that the system have adequate relative stability.

In this section, we shall show that the Nyquist plot indicates not only whether a system is stable but also the degree of stability of a stable system. The Nyquist plot also gives information as to how stability may be improved, if this is necessary. (See Chapter 9.)

In the following discussion, we shall assume that the systems considered have unity feedback. Note that it is always possible to reduce a system with feedback elements to

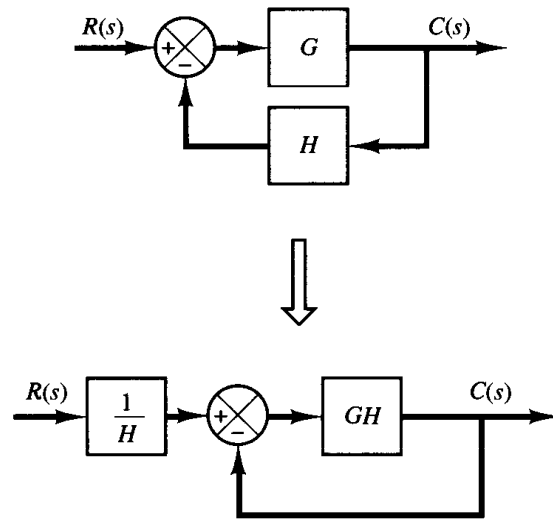


Figure 8-70
Modification of a system with feedback elements to a unity-feedback system.

a unity-feedback system, as shown in Figure 8-70. Hence, the extension of relative stability analysis for the unity-feedback system to nonunity-feedback systems is possible.

We shall also assume that, unless otherwise stated, the systems are minimum-phase systems; that is, the open-loop transfer function $G(s)$ has neither poles nor zeros in the right-half s plane.

Relative stability analysis by conformal mapping. One of the important problems in analyzing a control system is to find all closed-loop poles or at least those closest to the $j\omega$ axis (or the dominant pair of closed-loop poles). If the open-loop frequency-response characteristics of a system are known, it may be possible to estimate the closed-loop poles closest to the $j\omega$ axis. It is noted that the Nyquist locus $G(j\omega)$ need not be an analytically known function of ω . The entire Nyquist locus may be experimentally obtained. The technique to be presented here is essentially graphical and is based on a conformal mapping of the s plane into the $G(s)$ plane.

Consider the conformal mapping of constant- σ lines (lines $s = \sigma + j\omega$, where σ is constant and ω varies) and constant- ω lines (lines $s = \sigma + j\omega$, where ω is constant and σ varies) in the s plane. The $\sigma = 0$ line (the $j\omega$ axis) in the s plane maps into the Nyquist plot in the $G(s)$ plane. The constant- σ lines in the s plane map into curves that are similar to the Nyquist plot and are in a sense parallel to the Nyquist plot, as shown in Figure 8-71. The constant- ω lines in the s plane map into curves, also shown in Figure 8-71.

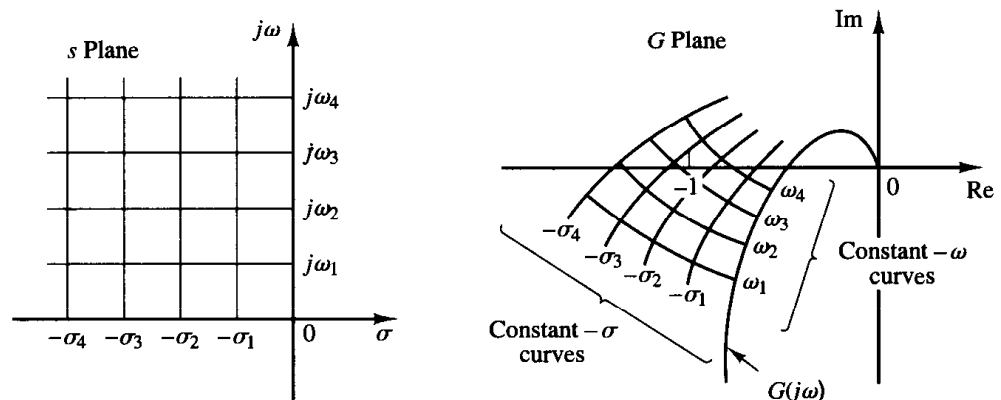
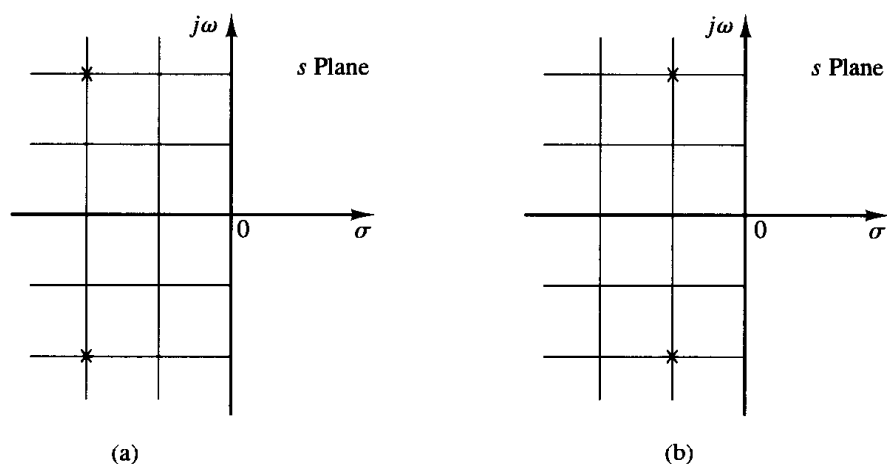


Figure 8-71
Conformal mapping of s -plane grids into the $G(s)$ plane.

Figure 8–72
Two systems with
two closed-loop
poles.



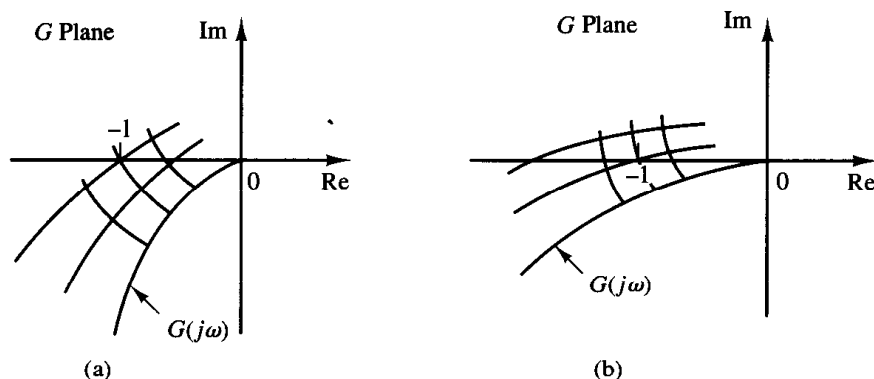
Although the shapes of constant- σ and constant- ω loci in the $G(s)$ plane and the closeness of approach of the $G(j\omega)$ locus to the $-1 + j0$ point depend on a particular $G(s)$, the closeness of approach of the $G(j\omega)$ locus to the $-1 + j0$ point is an indication of the relative stability of a stable system. In general, we may expect that the closer the $G(j\omega)$ locus is to the $-1 + j0$ point, the larger the maximum overshoot is in the step transient response and the longer it takes to damp out.

Consider the two systems shown in Figure 8–72(a) and (b). (In Figure 8–72, the $\times$'s indicate closed-loop poles.) System (a) is obviously more stable than system (b) because the closed-loop poles of system (a) are located farther left than those of system (b). Figures 8–73(a) and (b) show the conformal mapping of s -plane grids into the $G(s)$ plane. The closer the closed-loop poles are located to the $j\omega$ axis, the closer the $G(j\omega)$ locus is to the $-1 + j0$ point.

Phase and gain margins. Figure 8–74 shows the polar plots of $G(j\omega)$ for three different values of the open-loop gain K . For a large value of the gain K , the system is unstable. As the gain is decreased to a certain value, the $G(j\omega)$ locus passes through the $-1 + j0$ point. This means that with this gain value the system is on the verge of instability, and the system will exhibit sustained oscillations. For a small value of the gain K , the system is stable.

In general, the closer the $G(j\omega)$ locus comes to encircling the $-1 + j0$ point, the more oscillatory is the system response. The closeness of the $G(j\omega)$ locus to the $-1 + j0$ point can be used as a measure of the margin of stability. (This does not apply, however,

Figure 8–73
Conformal mappings
of s -plane grids for
the systems shown in
Figure 8–72 into the
 $G(s)$ plane.



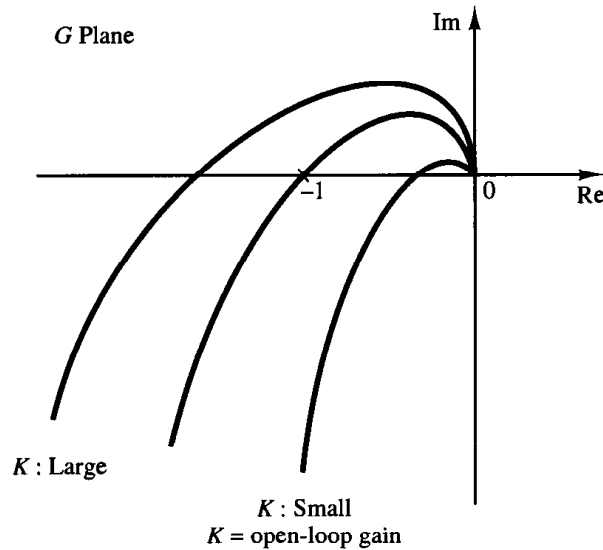


Figure 8-74

Polar plots of

$$\frac{K(1 + j\omega T_a)(1 + j\omega T_b) \cdots}{(j\omega)(1 + j\omega T_1)(1 + j\omega T_2) \cdots}$$

to conditionally stable systems.) It is common practice to represent the closeness in terms of phase margin and gain margin.

Phase margin: The phase margin is that amount of additional phase lag at the gain crossover frequency required to bring the system to the verge of instability. The gain crossover frequency is the frequency at which $|G(j\omega)|$, the magnitude of the open-loop transfer function, is unity. The phase margin γ is 180° plus the phase angle ϕ of the open-loop transfer function at the gain crossover frequency, or

$$\gamma = 180^\circ + \phi$$

Figures 8-75(a), (b), and (c) illustrate the phase margin of both a stable system and an unstable system in Bode diagrams, polar plots, and log-magnitude versus phase plots. In the polar plot, a line may be drawn from the origin to the point at which the unit circle crosses the $G(j\omega)$ locus. The angle from the negative real axis to this line is the phase margin. The phase margin is positive for $\gamma > 0$ and negative for $\gamma < 0$. For a minimum-phase system to be stable, the phase margin must be positive. In the logarithmic plots, the critical point in the complex plane corresponds to the 0 dB and -180° lines.

Gain margin: The gain margin is the reciprocal of the magnitude $|G(j\omega)|$ at the frequency at which the phase angle is -180° . Defining the phase crossover frequency ω_1 to be the frequency at which the phase angle of the open-loop transfer function equals -180° gives the gain margin K_g :

$$K_g = \frac{1}{|G(j\omega_1)|}$$

In terms of decibels,

$$K_g \text{ dB} = 20 \log K_g = -20 \log |G(j\omega_1)|$$

The gain margin expressed in decibels is positive if K_g is greater than unity and negative if K_g is smaller than unity. Thus, a positive gain margin (in decibels) means that the

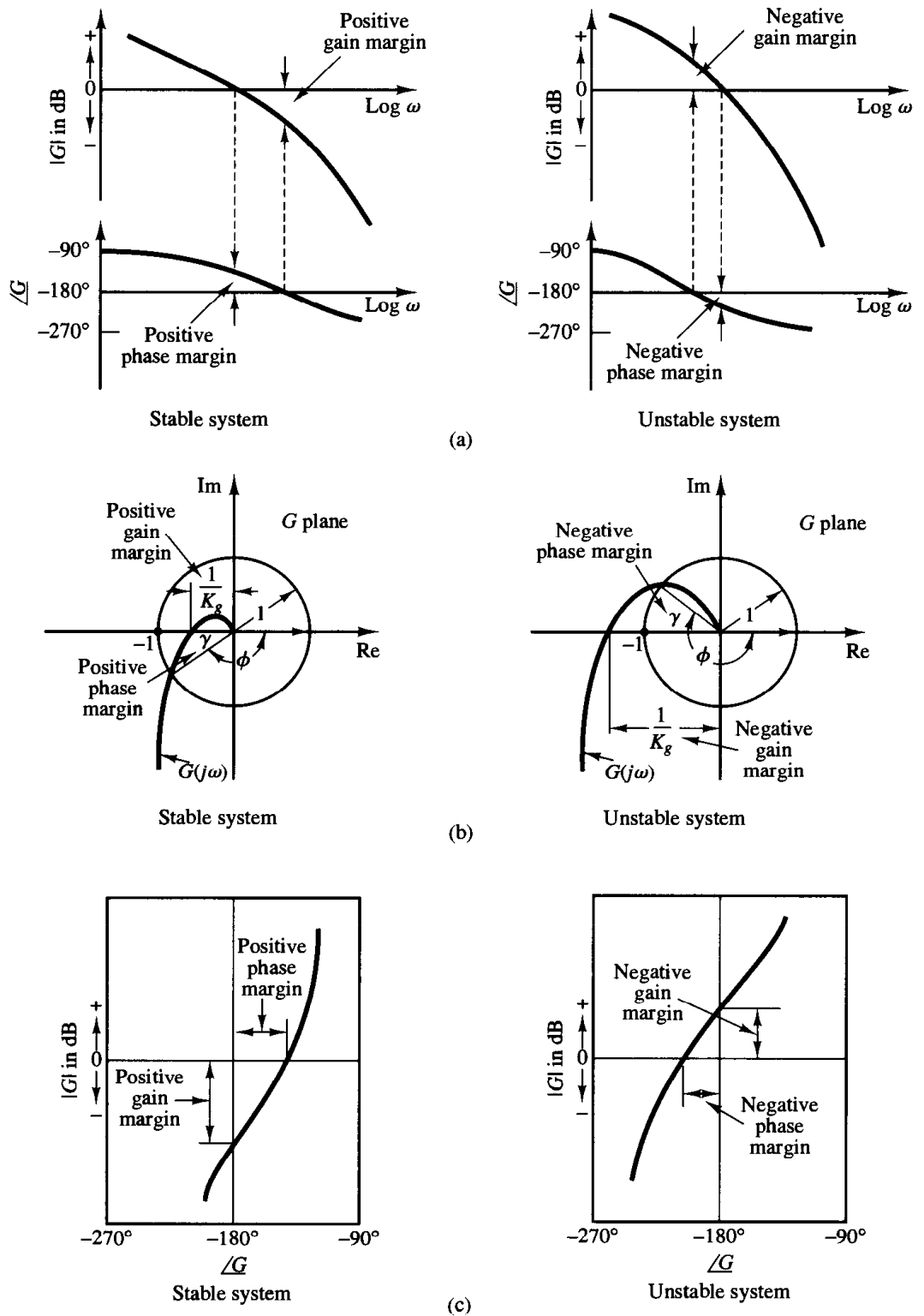


Figure 8-75
Phase and gain margins of stable and unstable systems.
(a) Bode diagrams;
(b) polar plots;
(c) log-magnitude versus phase plots.

system is stable, and a negative gain margin (in decibels) means that the system is unstable. The gain margin is shown in Figures 8-75(a), (b), and (c).

For a stable minimum-phase system, the gain margin indicates how much the gain can be increased before the system becomes unstable. For an unstable system, the gain margin is indicative of how much the gain must be decreased to make the system stable.

The gain margin of a first- or second-order system is infinite since the polar plots for such systems do not cross the negative real axis. Thus, theoretically, first- or second-order systems cannot be unstable. (Note, however, that so-called first- or second-order systems are only approximations in the sense that small time lags are neglected in deriving the system equations and are thus not truly first- or second-order systems. If these small lags are accounted for, the so-called first- or second-order systems may become unstable.)

It is noted that for a nonminimum-phase system with unstable open loop the stability condition will not be satisfied unless the $G(j\omega)$ plot encircles the $-1 + j0$ point. Hence, such a stable nonminimum-phase system will have negative phase and gain margins.

It is also important to point out that conditionally stable systems will have two or more phase crossover frequencies, and some higher-order systems with complicated numerator dynamics may also have two or more gain crossover frequencies, as shown in Figure 8-76. For stable systems having two or more gain crossover frequencies, the phase margin is measured at the highest gain crossover frequency.

A few comments on phase and gain margins. The phase and gain margins of a control system are a measure of the closeness of the polar plot to the $-1 + j0$ point. Therefore, these margins may be used as design criteria.

It should be noted that either the gain margin alone or the phase margin alone does not give a sufficient indication of the relative stability. Both should be given in the determination of relative stability.

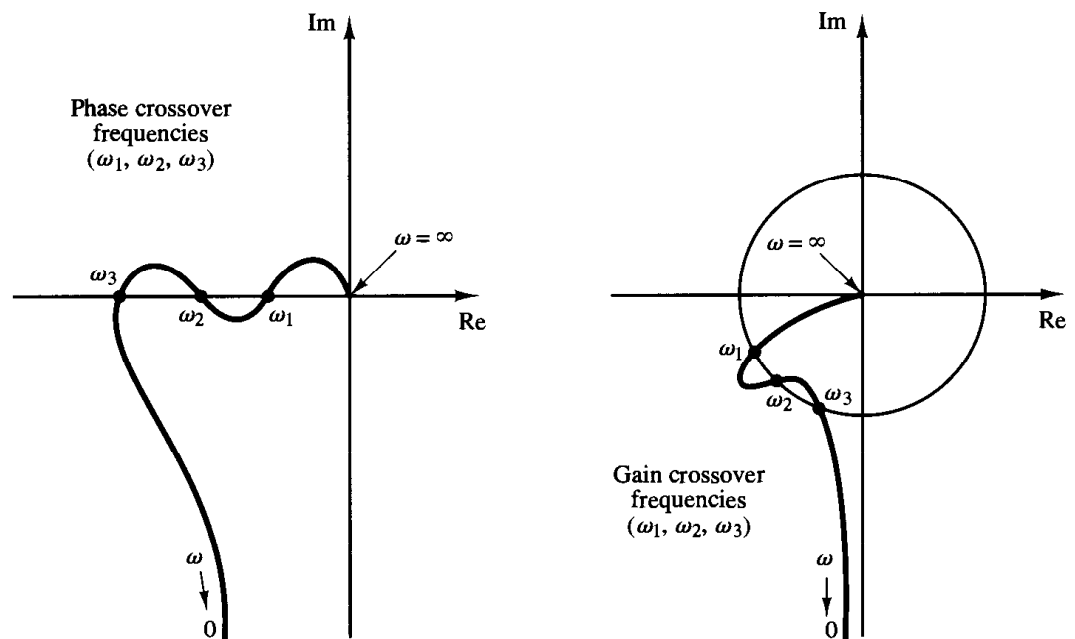


Figure 8-76
Polar plots showing more than two phase or gain crossover frequencies.

For a minimum-phase system, both the phase and gain margins must be positive for the system to be stable. Negative margins indicate instability.

Proper phase and gain margins ensure us against variations in the system components and are specified for definite values of frequency. The two values bound the behavior of the closed-loop system near the resonant frequency. For satisfactory performance, the phase margin should be between 30° and 60° , and the gain margin should be greater than 6 dB. With these values, a minimum-phase system has guaranteed stability, even if the open-loop gain and time constants of the components vary to a certain extent. Although the phase and gain margins give only rough estimates of the effective damping ratio of the closed-loop system, they do offer a convenient means for designing control systems or adjusting the gain constants of systems.

For minimum-phase systems, the magnitude and phase characteristics of the open-loop transfer function are definitely related. The requirement that the phase margin be between 30° and 60° means that in a Bode diagram the slope of the log-magnitude curve at the gain crossover frequency should be more gradual than -40 dB/decade. In most practical cases, a slope of -20 dB/decade is desirable at the gain crossover frequency for stability. If it is -40 dB/decade, the system could be either stable or unstable. (Even if the system is stable, however, the phase margin is small.) If the slope at the gain crossover frequency is -60 dB/decade or steeper, the system is most likely unstable.

EXAMPLE 8-20

Obtain the phase and gain margins of the system shown in Figure 8-77 for the two cases where $K = 10$ and $K = 100$.

The phase and gain margins can easily be obtained from the Bode diagram. A Bode diagram of the given open-loop transfer function with $K = 10$ is shown in Figure 8-78(a). The phase and gain margins for $K = 10$ are

$$\text{Phase margin} = 21^\circ, \quad \text{Gain margin} = 8 \text{ dB}$$

Therefore, the system gain may be increased by 8 dB before the instability occurs.

Increasing the gain from $K = 10$ to $K = 100$ shifts the 0-dB axis down by 20 dB, as shown in Figure 8-78(b). The phase and gain margins are

$$\text{Phase margin} = -30^\circ, \quad \text{Gain margin} = -12 \text{ dB}$$

Thus, the system is stable for $K = 10$ but unstable for $K = 100$.

Notice that one of the very convenient aspects of the Bode diagram approach is the ease with which the effects of gain changes can be evaluated.

Note that to obtain satisfactory performance we must increase the phase margin to $30^\circ \sim 60^\circ$. This can be done by decreasing the gain K . Decreasing K is not desirable, however, since a small value of K will yield a large error for the ramp input. This suggests that reshaping of the open-loop frequency-response curve by adding compensation may be necessary. Compensation techniques are discussed in detail in Chapter 9.

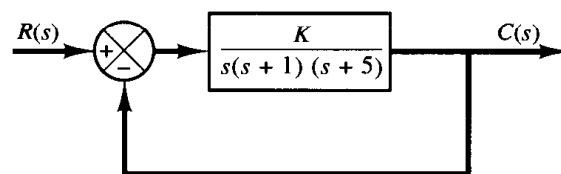
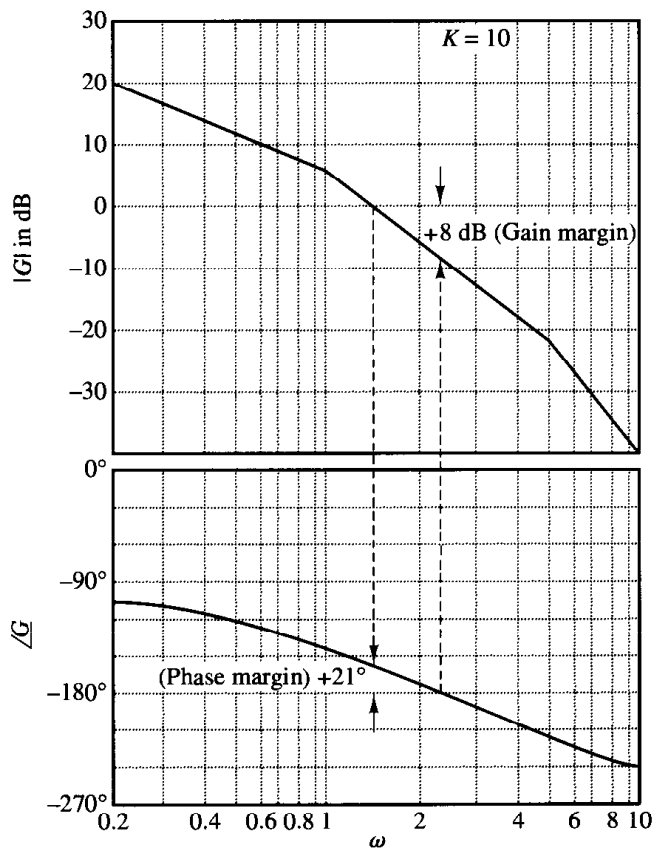
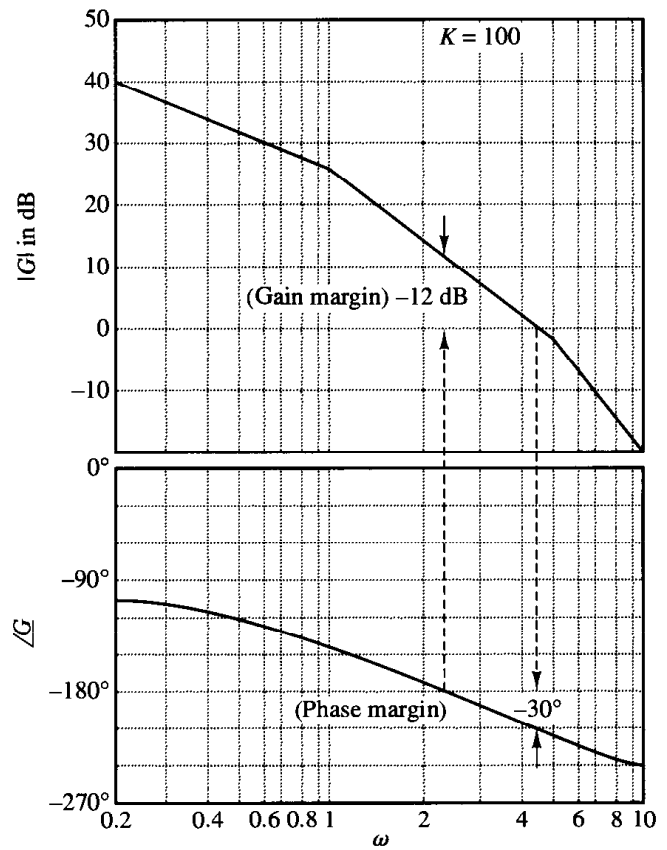


Figure 8-77
Control system.



(a)



(b)

Figure 8-78

Bode diagrams of the system shown in Figure 8-77 (a) with $K = 10$ and (b) with $K = 100$.

Resonant peak magnitude M_r and resonant peak frequency ω_r . Consider the system shown in Figure 8-79. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (8-9)$$

where ζ and ω_n are the damping ratio and the undamped natural frequency, respectively. The closed-loop frequency response is

$$\frac{C(j\omega)}{R(j\omega)} = \frac{1}{\left(1 - \frac{\omega^2}{\omega_n^2}\right) + j2\zeta \frac{\omega}{\omega_n}} = Me^{j\alpha}$$

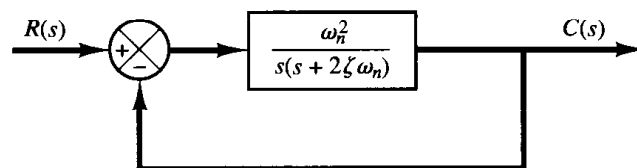


Figure 8-79
Control system.